

Logarithms

Worked Solutions with TikZ Graphs

Wisest Maths — Year 1 A-Level, Exponentials and Logarithms (ref **e12**)

This document contains all 70 *Logarithms* questions, each with a fully worked solution and a TikZ graph. Each diagram plots the relevant logarithmic curve $y = \log_b x$ (or its inverse exponential): the x -intercept $(1, 0)$ and key points are marked, the vertical asymptote $x = 0$ is shown, a dashed reference line passes through the solution for equation questions, and the line $y = x$ is added for inverse-function pairs.

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Logarithms 01 (el2-001)

Difficulty: Foundation / Marks: 1

Write the following using logarithm notation: $3^4 = 81$.

Worked solution.

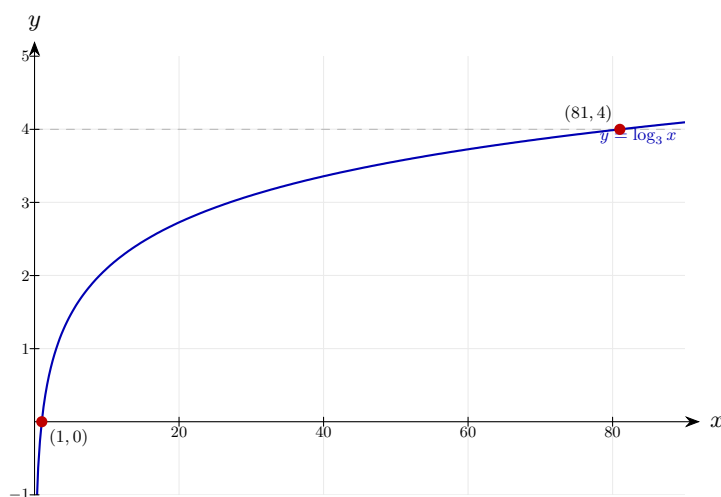
Use the rule: if $a^c = b$ then $\log_a b = c$.

$$3^4 = 81 \Rightarrow \log_3 81 = 4$$

The base is 3, the power is 4, and the result is 81.

Final answer: $\log_3 81 = 4$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 02 (e12-002)

Difficulty: Foundation / Marks: 1

Write the following using logarithm notation: $5^3 = 125$.

Worked solution.

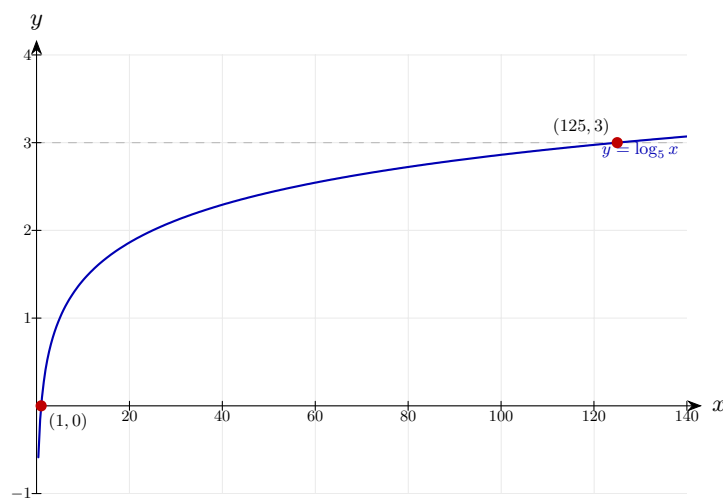
Apply $a^c = b \Leftrightarrow \log_a b = c$.

$$5^3 = 125 \Rightarrow \log_5 125 = 3$$

Base 5, power 3, result 125.

Final answer: $\log_5 125 = 3$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 03 (e12-003)

Difficulty: Foundation / Marks: 3

Write the following using logarithm notation: $7^2 = 49$, $4^{-2} = \frac{1}{16}$, and $27^{\frac{1}{3}} = 3$.

Worked solution.

Step 1: Convert $7^2 = 49$.

$$\log_7 49 = 2$$

Base 7, power 2.

Step 2: Convert $4^{-2} = \frac{1}{16}$.

$$\log_4 \frac{1}{16} = -2$$

The negative power gives a fractional result.

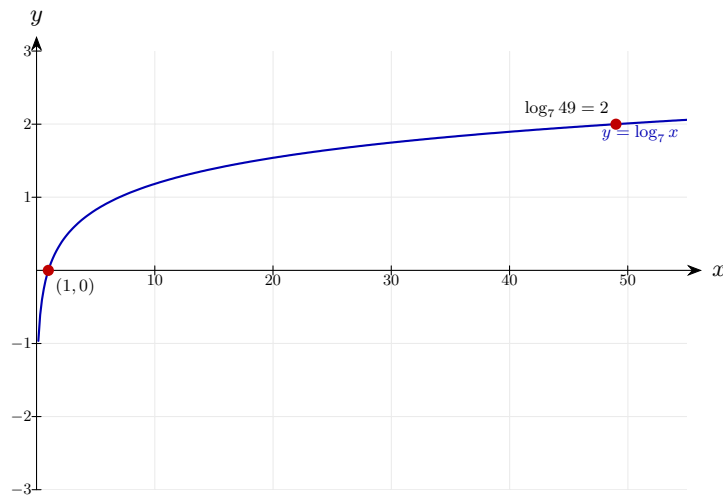
Step 3: Convert $27^{\frac{1}{3}} = 3$.

$$\log_{27} 3 = \frac{1}{3}$$

A power of $\frac{1}{3}$ is a cube root.

Final answer: $\log_7 49 = 2$; $\log_4 \frac{1}{16} = -2$; $\log_{27} 3 = \frac{1}{3}$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 04 (e12-004)

Difficulty: Foundation / Marks: 3

Write the following using index notation (do not evaluate): $\log_4 m = 3$, $\ln p = 7$, and $\log k = 5$.

Worked solution.

Step 1: Convert $\log_4 m = 3$.

$$4^3 = m$$

Base 4 raised to power 3 equals m .

Step 2: Convert $\ln p = 7$. Recall $\ln$ means $\log_e$.

$$e^7 = p$$

The natural log has base e .

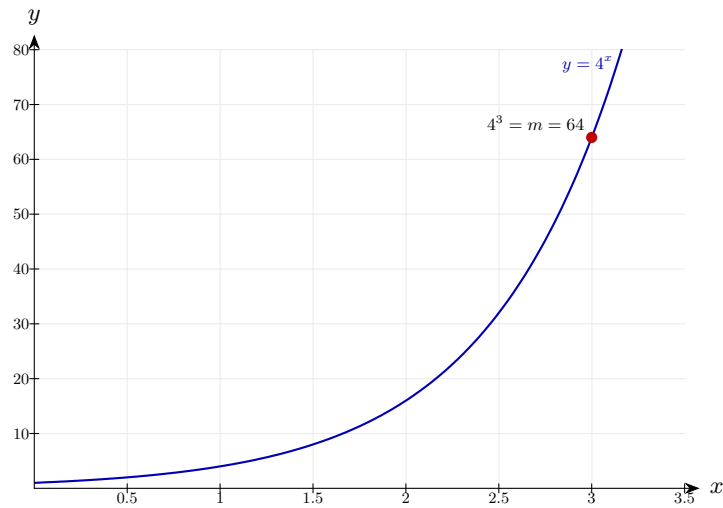
Step 3: Convert $\log k = 5$. Recall $\log$ without a base means $\log_{10}$.

$$10^5 = k$$

Base 10 is assumed when no base is written.

Final answer: $4^3 = m$; $e^7 = p$; $10^5 = k$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 05 (e12-005)

Difficulty: Foundation / Marks: 3

Write the following using index notation: $\log_a b = n$, $\ln q = r$, and $\log_6 w = z$.

Worked solution.

Step 1: $\log_a b = n$.

$$a^n = b$$

General form of the log-to-index conversion.

Step 2: $\ln q = r$.

$$e^r = q$$

$\ln$ has base e .

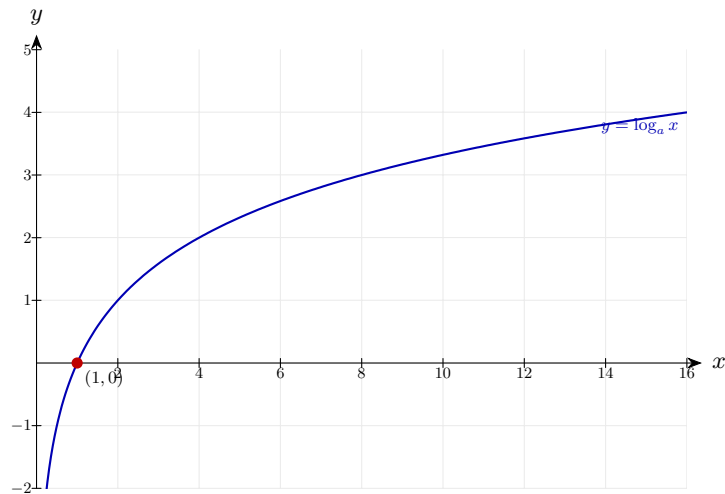
Step 3: $\log_6 w = z$.

$$6^z = w$$

Base 6 raised to z gives w .

Final answer: $a^n = b$; $e^r = q$; $6^z = w$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 06 (e12-006)

Difficulty: Foundation / Marks: 4

Find the value of each of the following:

(a) $\log_2 32$ (b) $\log_3 81$ (c) $\log_5 5$ (d) $\log_7 1$

Worked solution.

Step 1: Part (a): What power do you raise 2 to in order to get 32?

$$2^5 = 32 \Rightarrow \log_2 32 = 5$$

$$2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 16, 2^5 = 32.$$

Step 2: Part (b): $3^4 = 81$.

$$\log_3 81 = 4$$

$$3^4 = 81.$$

Step 3: Part (c): $\log_a a = 1$ for any valid base.

$$\log_5 5 = 1$$

Any number raised to the power 1 equals itself.

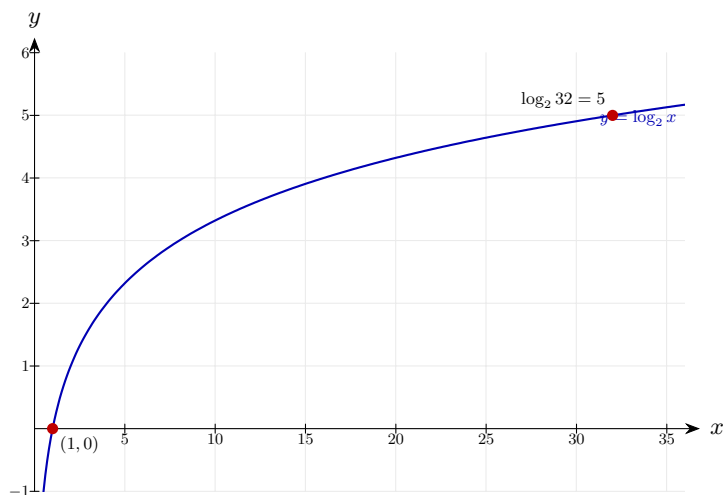
Step 4: Part (d): $\log_a 1 = 0$ for any valid base.

$$\log_7 1 = 0$$

$$7^0 = 1.$$

Final answer: (a)5 (b)4 (c)1 (d)0

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 07 (e12-007)

Difficulty: Foundation / Marks: 4

Find the value of each of the following:

(a) $\log_4 64$ (b) $\log_6 36$ (c) $\log_3 \frac{1}{9}$ (d) $\log_2 \frac{1}{8}$

Worked solution.

Step 1: Part (a): $4^3 = 64$.

$$\log_4 64 = 3$$

$$4^3 = 64.$$

Step 2: Part (b): $6^2 = 36$.

$$\log_6 36 = 2$$

$$6^2 = 36.$$

Step 3: Part (c): $\frac{1}{9} = 3^{-2}$.

$$\log_3 \frac{1}{9} = -2$$

A negative logarithm comes from a reciprocal.

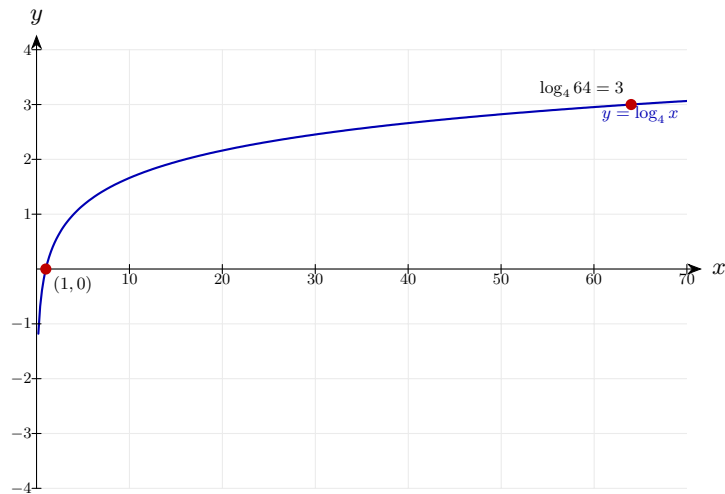
Step 4: Part (d): $\frac{1}{8} = 2^{-3}$.

$$\log_2 \frac{1}{8} = -3$$

$$2^{-3} = \frac{1}{8}.$$

Final answer: (a) 3 (b) 2 (c) -2 (d) -3

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 08 (e12-008)

Difficulty: Foundation / Marks: 4

Find the value of each of the following:

(a) $\log_9 3$ (b) $\log_8 2$ (c) $\log_{16} 4$ (d) $\log_{32} 2$

Worked solution.

Step 1: Part (a): $3 = 9^{1/2}$ (square root of 9).

$$\log_9 3 = \frac{1}{2}$$

$$9^{1/2} = \sqrt{9} = 3.$$

Step 2: Part (b): $2 = 8^{1/3}$ (cube root of 8).

$$\log_8 2 = \frac{1}{3}$$

$$8^{1/3} = \sqrt[3]{8} = 2.$$

Step 3: Part (c): $4 = 16^{1/2}$.

$$\log_{16} 4 = \frac{1}{2}$$

$$\sqrt{16} = 4.$$

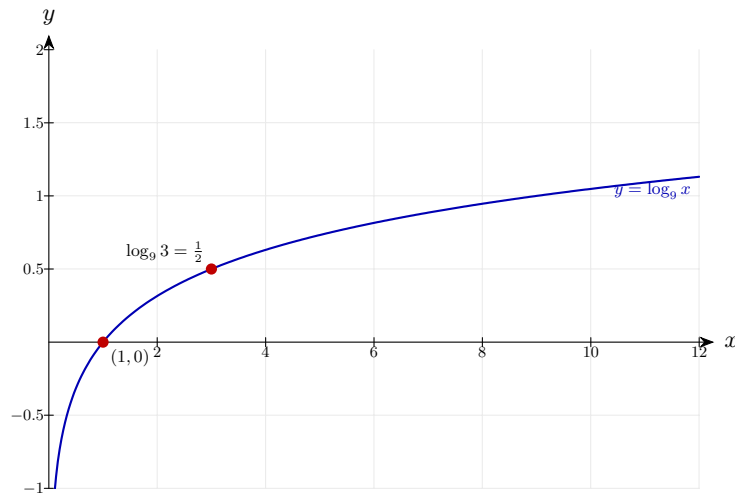
Step 4: Part (d): $2 = 32^{1/5}$ (fifth root of 32).

$$\log_{32} 2 = \frac{1}{5}$$

$$32^{1/5} = \sqrt[5]{32} = 2 \text{ since } 2^5 = 32.$$

Final answer: (a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{1}{5}$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 09 (e12-009)

Difficulty: Foundation / Marks: 4

Use a calculator to find each of the following, rounding to 3 decimal places:

(a) $\log 50$ (b) $\log 0.4$ (c) $\ln 5$ (d) $\ln 100$

Worked solution.

Step 1: Part (a): $\log 50 = \log_{10} 50$.

$$\log 50 = 1.699$$

$$10^{1.699} \approx 50.$$

Step 2: Part (b): $\log 0.4$ — result is negative since $0.4 < 1$.

$$\log 0.4 = -0.398$$

$$10^{-0.398} \approx 0.4.$$

Step 3: Part (c): $\ln 5$.

$$\ln 5 = 1.609$$

$$e^{1.609} \approx 5.$$

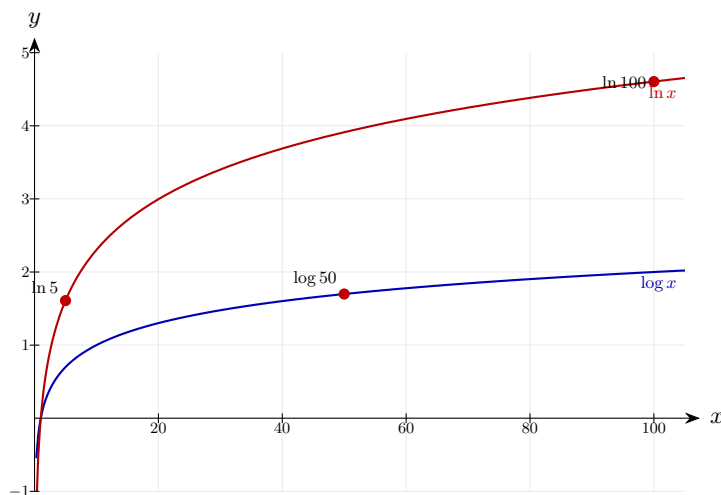
Step 4: Part (d): $\ln 100$.

$$\ln 100 = 4.605$$

$$e^{4.605} \approx 100.$$

Final answer: (a) 1.699 (b) -0.398 (c) 1.609 (d) 4.605

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 10 (e12-010)

Difficulty: Foundation / Marks: 3

Find the value of x (where $x > 0$) in each of the following:

(a) $\log_x 64 = 3$ (b) $\log_x 100 = 2$ (c) $\log_x 625 = 4$

Worked solution.

Step 1: Part (a): Convert to index form: $x^3 = 64$.

$$x = \sqrt[3]{64} = 4$$

$$4^3 = 64.$$

Step 2: Part (b): $x^2 = 100$.

$$x = \sqrt{100} = 10$$

$$10^2 = 100.$$

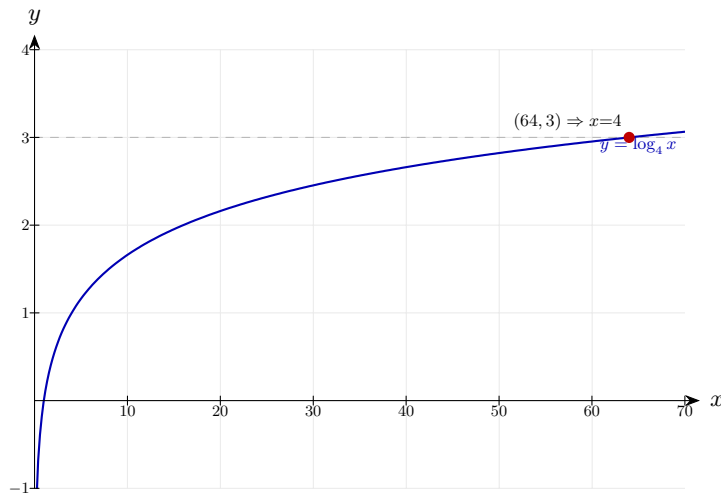
Step 3: Part (c): $x^4 = 625$.

$$x = \sqrt[4]{625} = 5$$

$$5^4 = 625.$$

Final answer: (a) $x = 4$ (b) $x = 10$ (c) $x = 5$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 11 (e12-011)

Difficulty: Foundation / Marks: 4

Find the value of x (where $x > 0$) by writing each equation in index notation:

(a) $\log_3 x = 4$ (b) $\log_5 x = 3$ (c) $\log_2 x = 6$ (d) $\log_4 x = -1$

Worked solution.

Step 1: Part (a): $\log_3 x = 4 \Rightarrow x = 3^4$.

$$x = 81$$

$$3^4 = 81.$$

Step 2: Part (b): $\log_5 x = 3 \Rightarrow x = 5^3$.

$$x = 125$$

$$5^3 = 125.$$

Step 3: Part (c): $\log_2 x = 6 \Rightarrow x = 2^6$.

$$x = 64$$

$$2^6 = 64.$$

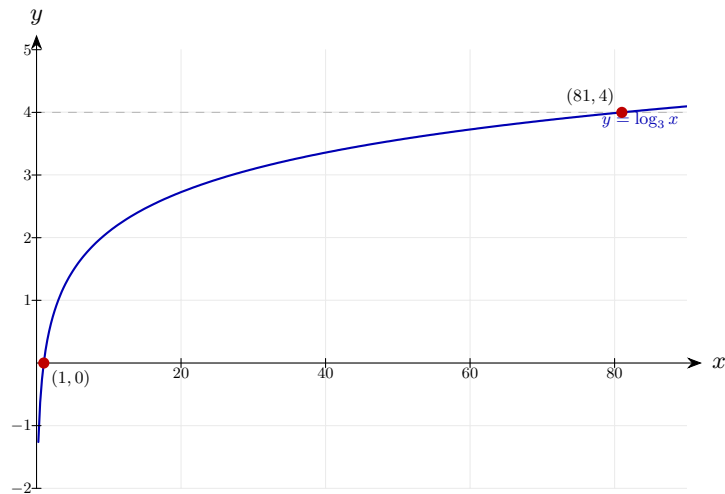
Step 4: Part (d): $\log_4 x = -1 \Rightarrow x = 4^{-1}$.

$$x = \frac{1}{4}$$

$$4^{-1} = \frac{1}{4}.$$

Final answer: (a) 81 (b) 125 (c) 64 (d) $\frac{1}{4}$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 12 (e12-012)

Difficulty: Foundation / Marks: 2

Without a calculator, explain why $\log_5 1 = 0$ and $\log_5 5 = 1$.

Worked solution.

Step 1: $\log_5 1 = 0$ because $5^0 = 1$.

$$5^0 = 1 \Rightarrow \log_5 1 = 0$$

Any positive base raised to the power zero gives 1.

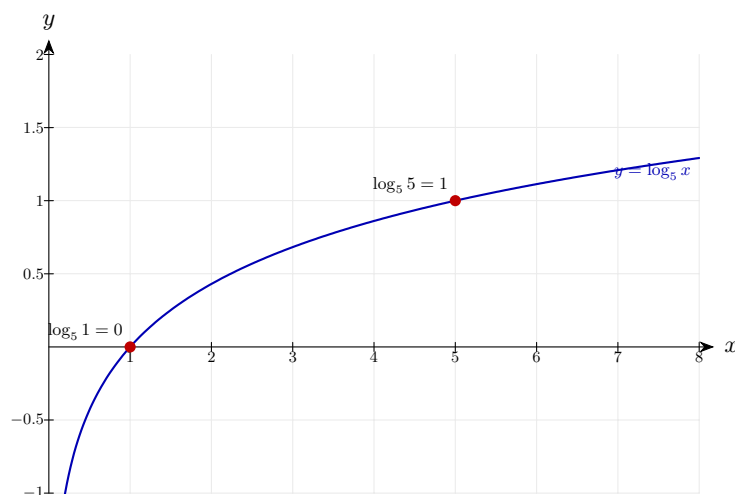
Step 2: $\log_5 5 = 1$ because $5^1 = 5$.

$$5^1 = 5 \Rightarrow \log_5 5 = 1$$

Any base raised to the power 1 equals itself.

Final answer: $\log_5 1 = 0$ (since $5^0 = 1$) and $\log_5 5 = 1$ (since $5^1 = 5$).

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 13 (e12-013)

Difficulty: Foundation / Marks: 3

State the values of: $\ln e$, $\ln 1$, and $\ln e^5$.

Worked solution.

Step 1: $\ln e = \log_e e = 1$.

$$\ln e = 1$$

Because $e^1 = e$.

Step 2: $\ln 1 = 0$.

$$\ln 1 = 0$$

Because $e^0 = 1$.

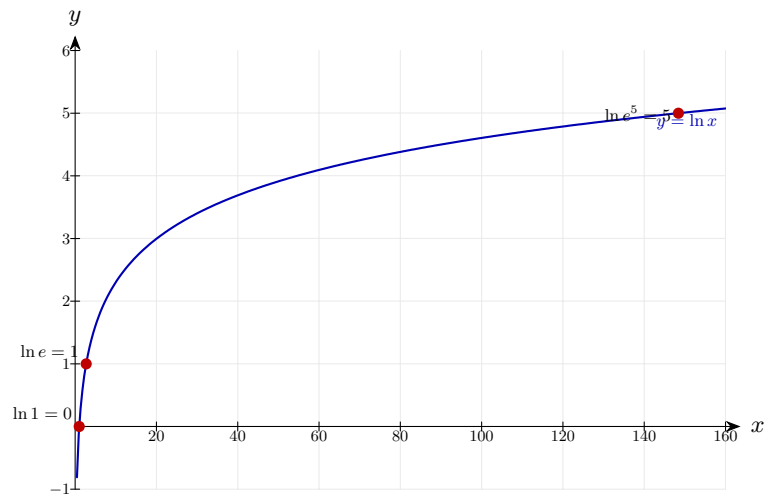
Step 3: $\ln e^5 = 5$.

$$\ln e^5 = 5$$

Using the inverse property: $\ln(e^x) = x$.

Final answer: $\ln e = 1$; $\ln 1 = 0$; $\ln e^5 = 5$.

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 14 (e12-014)

Difficulty: Foundation / Marks: 2

Solve $e^x = 7$, giving (i) an exact solution and (ii) a solution correct to 3 significant figures.

Worked solution.

Step 1: Apply $\ln$ to both sides.

$$\ln e^x = \ln 7 \Rightarrow x = \ln 7$$

$\ln$ and e are inverse functions, so $\ln e^x = x$.

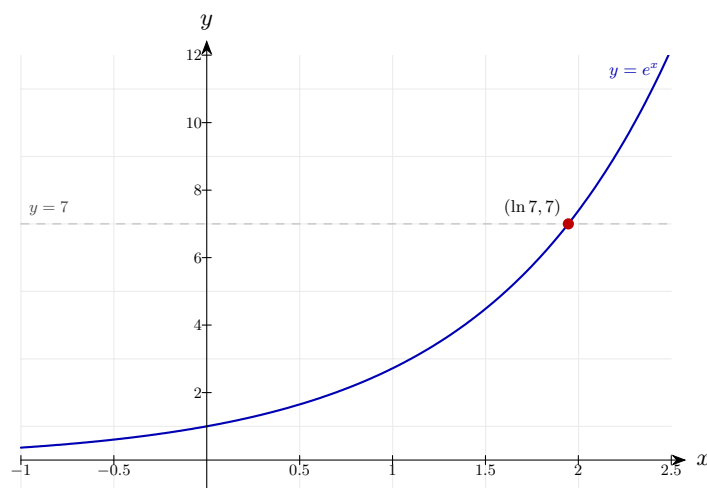
Step 2: Evaluate to 3 s.f.

$$x = \ln 7 \approx 1.95$$

Using a calculator.

Final answer: (i) $x = \ln 7$ (ii) $x \approx 1.95$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 15 (e12-015)

Difficulty: Foundation / Marks: 6

Solve each equation, giving (i) an exact solution and (ii) a solution to 3 s.f.:

(a) $e^x = 15$ (b) $e^x = 0.3$ (c) $3e^x = 24$

Worked solution.

Step 1: Part (a): Apply $\ln$ to both sides.

$$x = \ln 15 \approx 2.71$$

Direct application of $\ln$.

Step 2: Part (b): $e^x = 0.3$. Since $0.3 < 1$, the answer is negative.

$$x = \ln 0.3 \approx -1.20$$

$\ln$ of a number between 0 and 1 is negative.

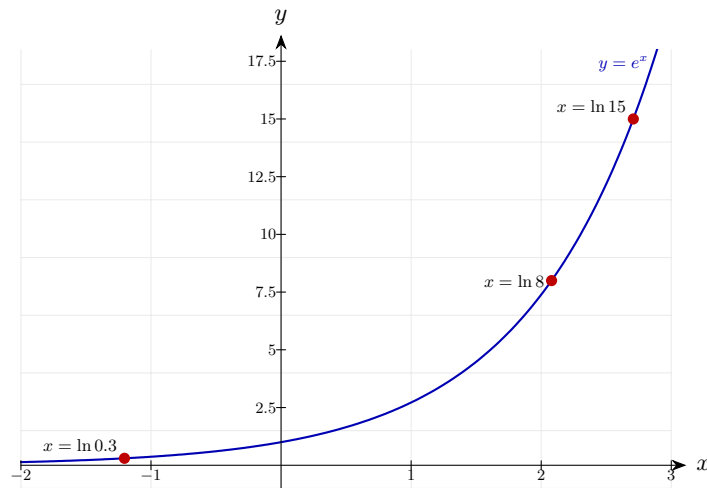
Step 3: Part (c): Divide by 3 first.

$$e^x = 8 \Rightarrow x = \ln 8 \approx 2.08$$

Isolate e^x before taking $\ln$.

Final answer: (a) $x = \ln 15 \approx 2.71$ (b) $x = \ln 0.3 \approx -1.20$ (c) $x = \ln 8 \approx 2.08$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 16 (e12-016)

Difficulty: Foundation / Marks: 6

Solve each equation, giving (i) an exact solution and (ii) a solution to 3 s.f.:

(a) $e^x = 430$ (b) $4e^x = 36$ (c) $e^x = 0.05$

Worked solution.

Step 1: Part (a): $x = \ln 430$.

$$x = \ln 430 \approx 6.06$$

Direct application.

Step 2: Part (b): Divide both sides by 4 to get $e^x = 9$.

$$x = \ln 9 \approx 2.20$$

Rearrange before applying $\ln$.

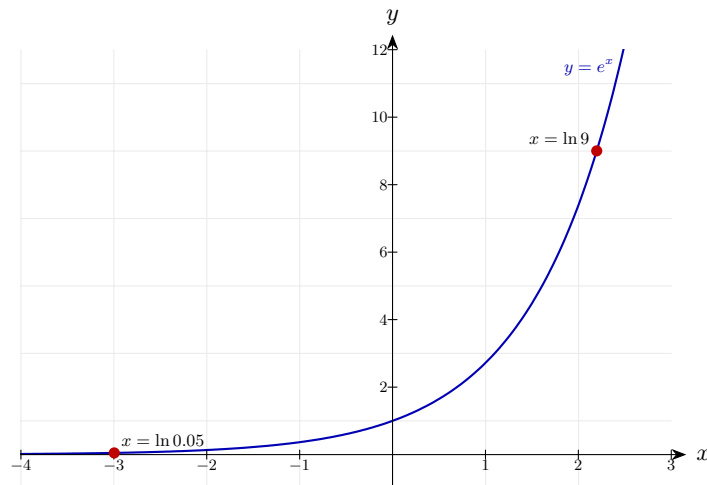
Step 3: Part (c): $e^x = 0.05$. Result is negative.

$$x = \ln 0.05 \approx -3.00$$

$$\ln(0.05) = \ln\left(\frac{1}{20}\right) \approx -2.996.$$

Final answer: (a) $x = \ln 430 \approx 6.06$ (b) $x = \ln 9 \approx 2.20$ (c) $x = \ln 0.05 \approx -3.00$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 17 (e12-017)

Difficulty: Foundation / Marks: 2

Solve $\ln x = 4$, giving (i) an exact solution and (ii) a solution to 3 s.f.

Worked solution.

Step 1: Apply e to both sides to undo the $\ln$.

$$e^{\ln x} = e^4 \Rightarrow x = e^4$$

e and $\ln$ are inverse functions.

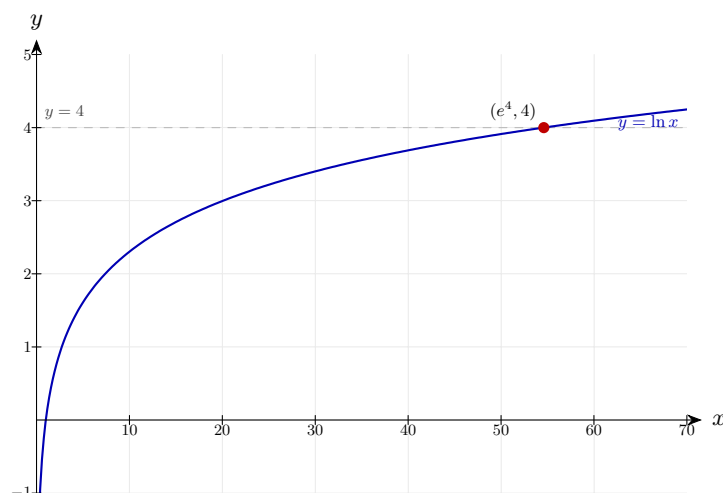
Step 2: Evaluate to 3 s.f.

$$x = e^4 \approx 54.6$$

$$e^4 \approx 54.598.$$

Final answer: (i) $x = e^4$ (ii) $x \approx 54.6$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 18 (e12-018)

Difficulty: Foundation / Marks: 6

Solve each equation, giving (i) an exact solution and (ii) a solution to 3 s.f.:

(a) $\ln x = 6$ (b) $\ln x = 0.8$ (c) $4 \ln x = 20$

Worked solution.

Step 1: Part (a): Raise e to both sides.

$$x = e^6 \approx 403$$

$$e^6 \approx 403.4.$$

Step 2: Part (b): $x = e^{0.8}$.

$$x = e^{0.8} \approx 2.23$$

$$e^{0.8} \approx 2.2255.$$

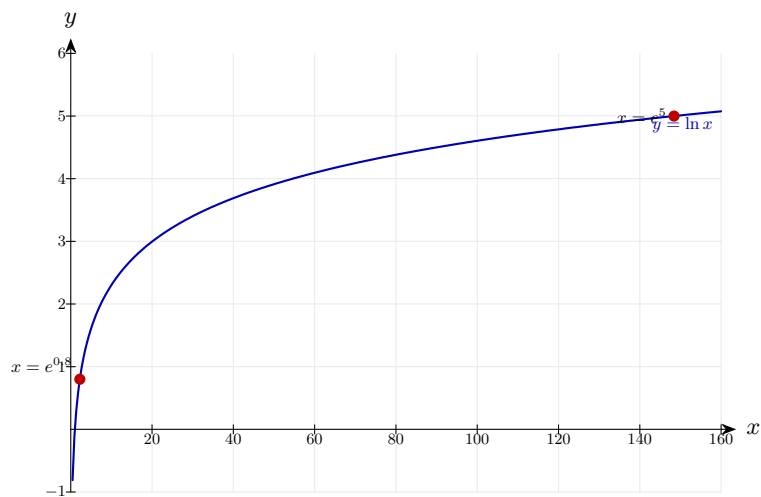
Step 3: Part (c): Divide by 4 first to get $\ln x = 5$.

$$x = e^5 \approx 148$$

$$e^5 \approx 148.4.$$

Final answer: (a) $x = e^6 \approx 403$ (b) $x = e^{0.8} \approx 2.23$ (c) $x = e^5 \approx 148$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 19 (e12-019)

Difficulty: Foundation / Marks: 6

Solve each equation, giving (i) an exact solution and (ii) a solution to 3 s.f.:

(a) $\ln x = -2$ (b) $5 \ln x = 30$ (c) $\ln 3x = 7$

Worked solution.

Step 1: Part (a): $\ln x = -2 \Rightarrow x = e^{-2}$.

$$x = e^{-2} = \frac{1}{e^2} \approx 0.135$$

A negative result for $\ln x$ means $0 < x < 1$.

Step 2: Part (b): $5 \ln x = 30 \Rightarrow \ln x = 6 \Rightarrow x = e^6$.

$$x = e^6 \approx 403$$

Divide by 5 first.

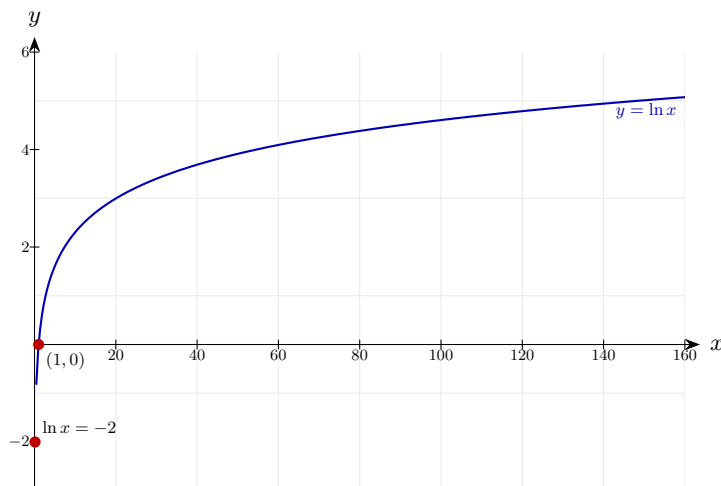
Step 3: Part (c): $\ln 3x = 7 \Rightarrow 3x = e^7 \Rightarrow x = \frac{e^7}{3}$.

$$x = \frac{e^7}{3} \approx 366$$

Raise e to both sides, then divide by 3.

Final answer: (a) $x = e^{-2} \approx 0.135$ (b) $x = e^6 \approx 403$ (c) $x = \frac{e^7}{3} \approx 366$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 20 (e12-020)

Difficulty: Foundation / Marks: 4

Solve each equation, giving (i) an exact solution and (ii) a solution to 3 s.f.:

(a) $\ln(x - 1) = 3$ (b) $\ln(2x + 1) = 5$

Worked solution.

Step 1: Part (a): Raise e to both sides.

$$x - 1 = e^3 \Rightarrow x = e^3 + 1 \approx 21.1$$

$e^3 \approx 20.09$, so $x \approx 21.1$.

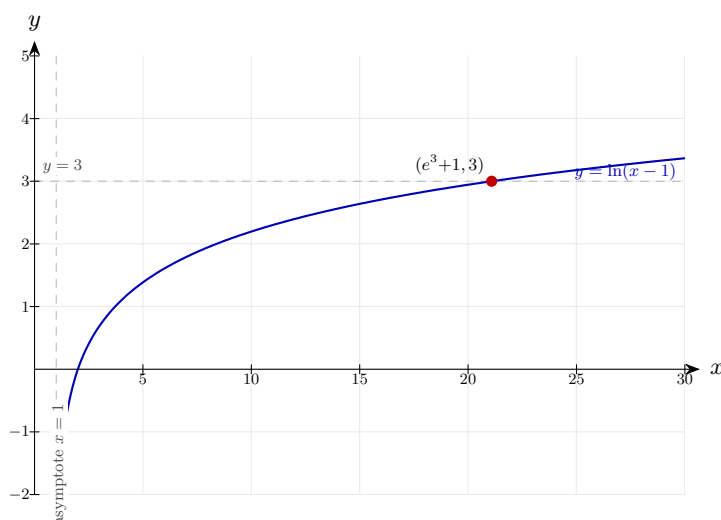
Step 2: Part (b): Raise e to both sides.

$$2x + 1 = e^5 \Rightarrow 2x = e^5 - 1 \Rightarrow x = \frac{e^5 - 1}{2} \approx 73.7$$

$e^5 \approx 148.41$, so $\frac{148.41-1}{2} \approx 73.7$.

Final answer: (a) $x = e^3 + 1 \approx 21.1$ (b) $x = \frac{e^5 - 1}{2} \approx 73.7$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 21 (e12-021)

Difficulty: Foundation / Marks: 3

Without a calculator, simplify:

(a) $e^{\ln 9}$ (b) $\ln(e^{12})$ (c) $e^{\ln x}$

Worked solution.

Step 1: Part (a): e and $\ln$ are inverse functions.

$$e^{\ln 9} = 9$$

$e^{\ln a} = a$ for any $a > 0$.

Step 2: Part (b): $\ln(e^{12}) = 12$.

$$\ln(e^{12}) = 12$$

$\ln(e^x) = x$.

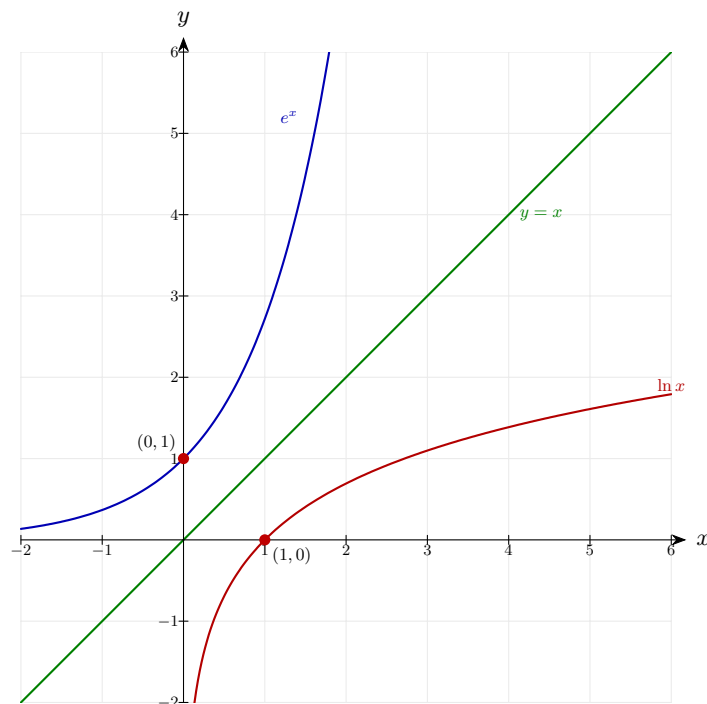
Step 3: Part (c): $e^{\ln x} = x$.

$$e^{\ln x} = x$$

The general inverse property.

Final answer: (a) 9 (b) 12 (c) x

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 22 (e12-022)

Difficulty: Foundation / Marks: 3

The graph of $y = \ln x$ is a reflection of $y = e^x$ in the line $y = x$.

(a) State the coordinates where $y = \ln x$ crosses the x -axis.

(b) State the equation of the asymptote of $y = \ln x$.

(c) State the domain of $y = \ln x$.

Worked solution.

Step 1: Part (a): $\ln x = 0 \Rightarrow x = e^0 = 1$.

$$(1, 0)$$

The graph crosses the x -axis at $x = 1$ since $\ln 1 = 0$.

Step 2: Part (b): As $x \rightarrow 0^+$, $\ln x \rightarrow -\infty$. The asymptote is the y -axis.

$$x = 0$$

The graph approaches but never touches $x = 0$.

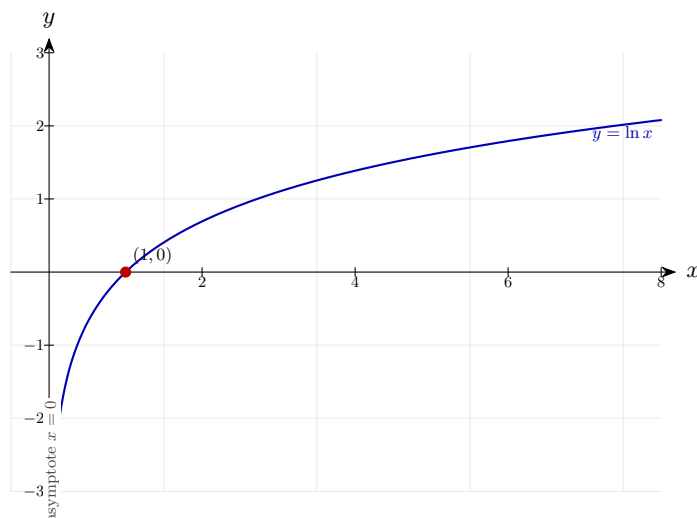
Step 3: Part (c): $\ln x$ is only defined for positive x .

$$x > 0$$

You cannot take the logarithm of zero or a negative number.

Final answer: (a) $(1, 0)$. (b) $x = 0$. (c) $x > 0$.

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 23 (e12-023)

Difficulty: Foundation / Marks: 3

Given that $\log_3 x = 2$ and $\log_3 y = 6$, write y in terms of x using index notation.

Worked solution.

Step 1: Convert $\log_3 x = 2$ to index form.

$$x = 3^2 \Rightarrow 3 = x^{1/2}$$

Express 3 in terms of x .

Step 2: Convert $\log_3 y = 6$ to index form.

$$y = 3^6$$

Express y using base 3.

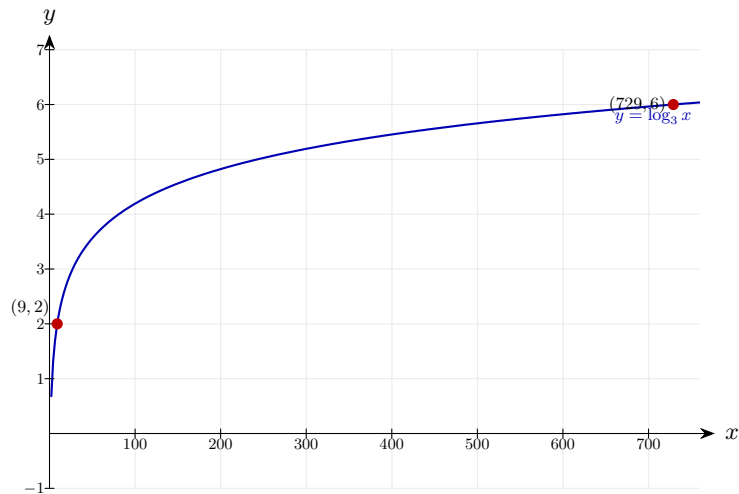
Step 3: Substitute $3 = x^{1/2}$ into $y = 3^6$.

$$y = \left(x^{1/2}\right)^6 = x^3$$

Use the laws of indices: $(x^{1/2})^6 = x^3$.

Final answer: $y = x^3$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 24 (e12-024)

Difficulty: Foundation / Marks: 3

Given that $\log_2 x = 4$ and $\log_2 y = 12$, write y in terms of x .

Worked solution.

Step 1: From $\log_2 x = 4$: $x = 2^4 \Rightarrow 2 = x^{1/4}$.

$$2 = x^{1/4}$$

Raise both sides to the power $\frac{1}{4}$.

Step 2: From $\log_2 y = 12$: $y = 2^{12}$.

$$y = 2^{12}$$

Index form.

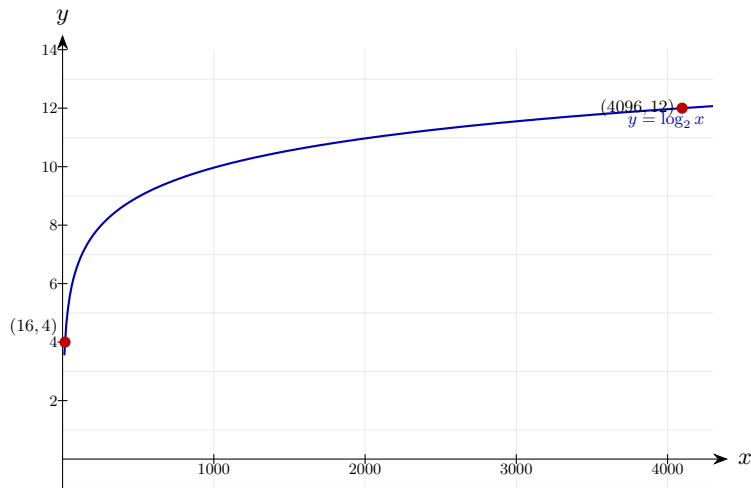
Step 3: Substitute $2 = x^{1/4}$.

$$y = \left(x^{1/4}\right)^{12} = x^3$$

$$\frac{12}{4} = 3.$$

Final answer: $y = x^3$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 25 (e12-025)

Difficulty: Foundation / Marks: 3

Given that $\ln x = 3$ and $\ln y = 9$, write y in terms of x .

Worked solution.

Step 1: From $\ln x = 3$: $x = e^3 \Rightarrow e = x^{1/3}$.

$$e = x^{1/3}$$

Raise both sides to $\frac{1}{3}$.

Step 2: From $\ln y = 9$: $y = e^9$.

$$y = e^9$$

Index form.

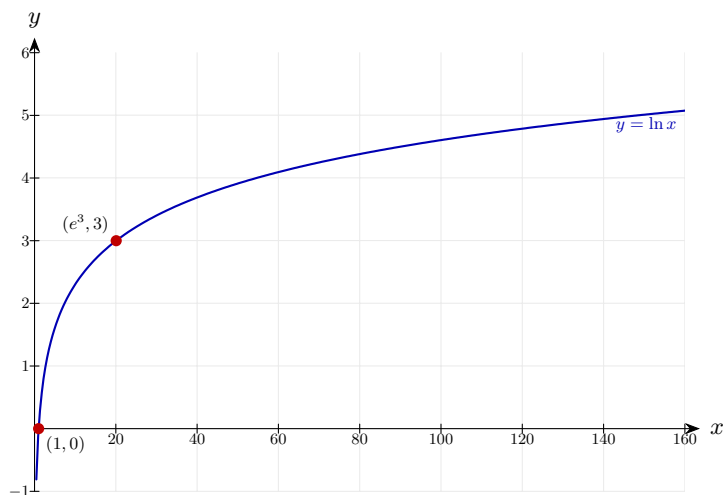
Step 3: Substitute $e = x^{1/3}$.

$$y = \left(x^{1/3}\right)^9 = x^3$$

$$\frac{9}{3} = 3.$$

Final answer: $y = x^3$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 26 (e12-026)

Difficulty: Foundation / Marks: 3

Given that $\ln x = 4$ and $\ln y = 2$, write y in terms of x .

Worked solution.

Step 1: From $\ln x = 4$: $x = e^4 \Rightarrow e = x^{1/4}$.

$$e = x^{1/4}$$

Raise both sides to $\frac{1}{4}$.

Step 2: From $\ln y = 2$: $y = e^2$.

$$y = e^2$$

Index form.

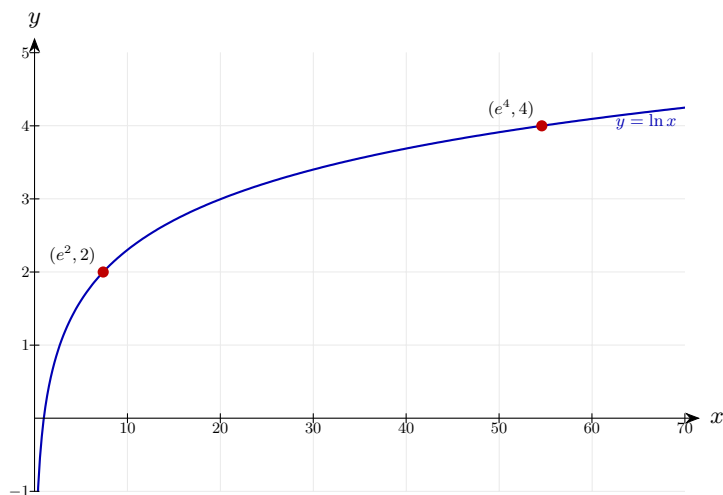
Step 3: Substitute $e = x^{1/4}$.

$$y = \left(x^{1/4}\right)^2 = x^{1/2} = \sqrt{x}$$

$$\frac{2}{4} = \frac{1}{2}.$$

Final answer: $y = \sqrt{x}$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 27 (e12-027)

Difficulty: Foundation / Marks: 3

Solve $7e^x = 35$. Give (i) an exact solution and (ii) a solution to 3 s.f.

Worked solution.

Step 1: Divide both sides by 7.

$$e^x = 5$$

Isolate the exponential term.

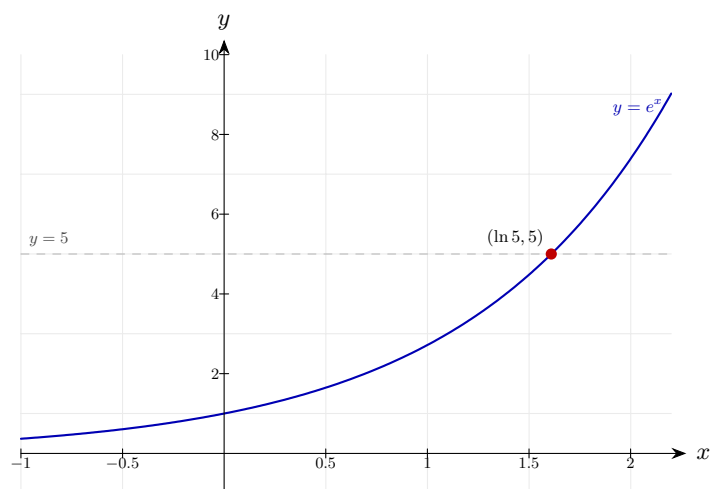
Step 2: Apply $\ln$ to both sides.

$$x = \ln 5 \approx 1.61$$

$\ln 5 \approx 1.609$.

Final answer: (i) $x = \ln 5$ (ii) $x \approx 1.61$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 28 (e12-028)

Difficulty: Foundation / Marks: 3

Solve $6 \ln x = 42$. Give (i) an exact solution and (ii) a solution to 3 s.f.

Worked solution.

Step 1: Divide both sides by 6.

$$\ln x = 7$$

Isolate the $\ln x$ term.

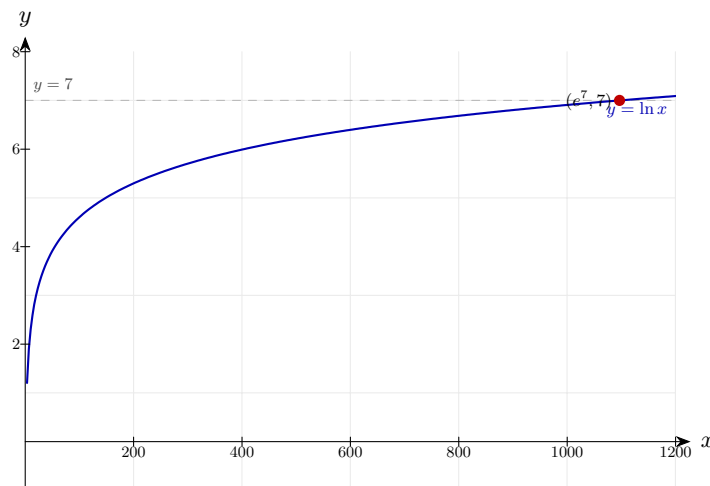
Step 2: Raise e to both sides.

$$x = e^7 \approx 1100$$

$e^7 \approx 1096.6$, which rounds to 1100 to 3 s.f.

Final answer: (i) $x = e^7$ (ii) $x \approx 1100$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 29 (e12-029)

Difficulty: Foundation / Marks: 3

Solve $\ln(5x) = 3$. Give (i) an exact solution and (ii) a solution to 3 s.f.

Worked solution.

Step 1: Raise e to both sides.

$$5x = e^3$$

$$e^{\ln(5x)} = 5x.$$

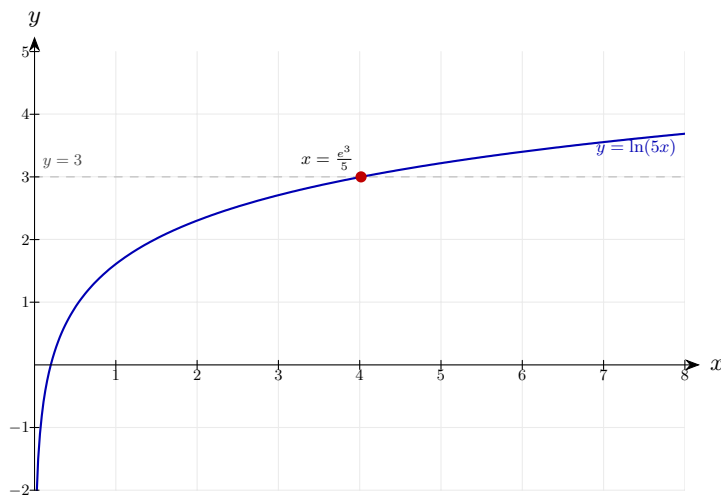
Step 2: Divide by 5.

$$x = \frac{e^3}{5} \approx 4.02$$

$$e^3 \approx 20.09 \Rightarrow \frac{20.09}{5} \approx 4.02.$$

Final answer: (i) $x = \frac{e^3}{5}$ (ii) $x \approx 4.02$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 30 (e12-030)

Difficulty: Foundation / Marks: 4

Solve $2e^{x+1} = 14$. Give (i) an exact solution and (ii) a solution to 3 s.f.

Worked solution.

Step 1: Divide both sides by 2.

$$e^{x+1} = 7$$

Isolate the exponential.

Step 2: Apply $\ln$.

$$x + 1 = \ln 7$$

$$\ln e^{x+1} = x + 1.$$

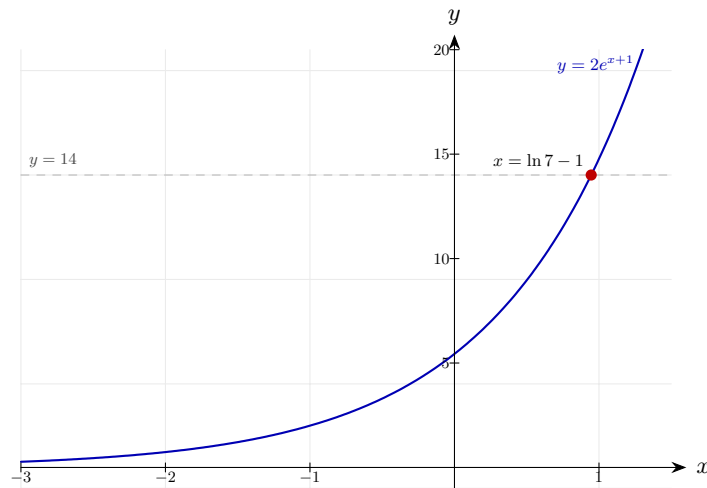
Step 3: Subtract 1.

$$x = \ln 7 - 1 \approx 0.946$$

$$\ln 7 \approx 1.946.$$

Final answer: (i) $x = \ln 7 - 1$ (ii) $x \approx 0.946$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 31 (e12-031)

Difficulty: Foundation / Marks: 5

- (a) Sketch the graph of $y = \ln x$, labelling the x -intercept and the asymptote.
- (b) On the same axes, sketch $y = \ln(x - 2)$. State its x -intercept and asymptote.

Worked solution.

Step 1: Part (a): Key features of $y = \ln x$.

x -intercept: $(1, 0)$; asymptote: $x = 0$

$\ln 1 = 0$; as $x \rightarrow 0^+$, $\ln x \rightarrow -\infty$.

Step 2: Part (b): $y = \ln(x - 2)$ is $y = \ln x$ shifted 2 units to the right.

x -intercept: set $\ln(x - 2) = 0 \Rightarrow x - 2 = 1 \Rightarrow x = 3 \Rightarrow (3, 0)$

The x -intercept shifts from 1 to 3.

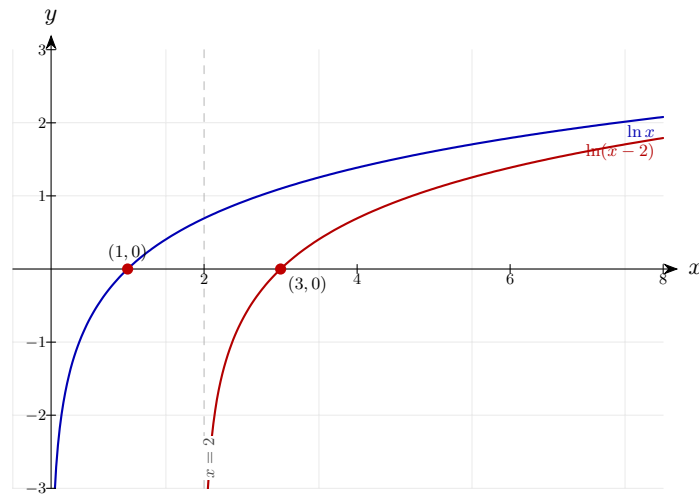
Step 3: The asymptote also shifts right by 2.

Asymptote: $x = 2$

$\ln(x - 2)$ is undefined when $x \leq 2$.

Final answer: (b) x -intercept $(3, 0)$; asymptote $x = 2$.

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 32 (e12-032)

Difficulty: Foundation / Marks: 5

The number of bacteria N in a culture after t hours satisfies $N = 200e^{0.3t}$.

(a) Find the initial number of bacteria.

(b) Find the exact time when the number of bacteria reaches 2000.

(c) Give the time to 3 s.f.

Worked solution.

Step 1: Part (a): Set $t = 0$.

$$N(0) = 200e^0 = 200$$

Initial count is 200 bacteria.

Step 2: Part (b): Set $N = 2000$ and solve for t .

$$200e^{0.3t} = 2000 \Rightarrow e^{0.3t} = 10$$

Divide by 200.

Step 3: Apply $\ln$ to both sides.

$$0.3t = \ln 10 \Rightarrow t = \frac{\ln 10}{0.3}$$

Exact solution.

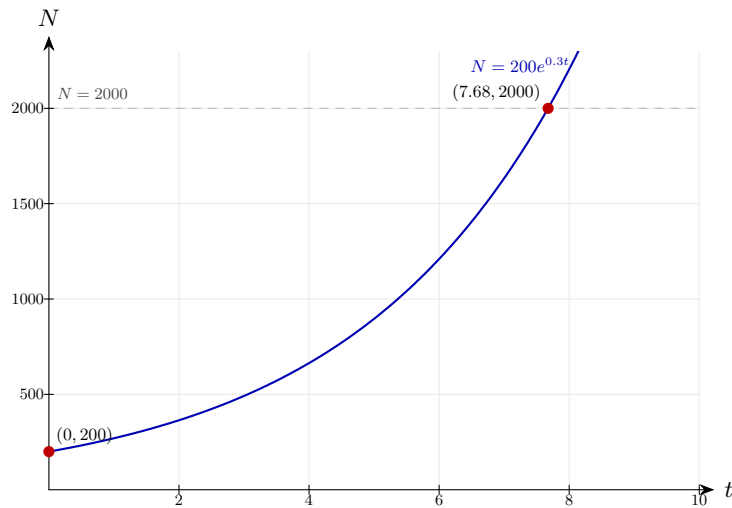
Step 4: Part (c): Evaluate to 3 s.f.

$$t = \frac{\ln 10}{0.3} \approx 7.68 \text{ hours}$$

$$\ln 10 \approx 2.303 \Rightarrow \frac{2.303}{0.3} \approx 7.68.$$

Final answer: (a) 200. (b) $t = \frac{\ln 10}{0.3}$. (c) $t \approx 7.68$ hours.

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 33 (e12-033)

Difficulty: Foundation / Marks: 5

A radioactive substance decays according to $M = 500e^{-0.04t}$, where M is the mass in grams and t is the time in years.

- (a) State the initial mass.
- (b) Find the exact time for the mass to halve.
- (c) Give the half-life to 3 s.f.

Worked solution.

Step 1: Part (a): Initial mass at $t = 0$.

$$M(0) = 500e^0 = 500 \text{ g}$$

The initial mass is 500 g.

Step 2: Part (b): Set $M = 250$ (half of 500).

$$500e^{-0.04t} = 250 \Rightarrow e^{-0.04t} = 0.5$$

Divide by 500.

Step 3: Apply $\ln$ to both sides.

$$-0.04t = \ln 0.5 = -\ln 2 \Rightarrow t = \frac{\ln 2}{0.04}$$

Divide by -0.04 . The half-life formula.

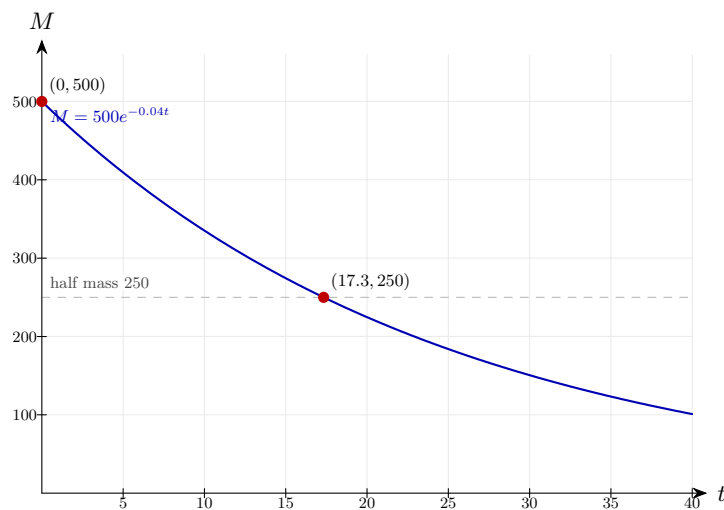
Step 4: Part (c): Evaluate.

$$t = \frac{\ln 2}{0.04} \approx 17.3 \text{ years}$$

$$\ln 2 \approx 0.6931 \Rightarrow \frac{0.6931}{0.04} \approx 17.3.$$

Final answer: (a) 500 g. (b) $t = \frac{\ln 2}{0.04}$. (c) $t \approx 17.3$ years.

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 34 (e12-034)

Difficulty: Foundation / Marks: 6

(a) Solve $e^{2x} = 50$, giving an exact answer and a decimal answer to 3 s.f.

(b) Solve $\ln(x^2) = 6$, giving exact answers.

Worked solution.

Step 1: Part (a): Apply $\ln$ to both sides.

$$2x = \ln 50 \Rightarrow x = \frac{\ln 50}{2}$$

Exact solution.

Step 2: Evaluate to 3 s.f.

$$x = \frac{\ln 50}{2} \approx \frac{3.912}{2} \approx 1.96$$

$\ln 50 \approx 3.912$.

Step 3: Part (b): $\ln(x^2) = 6 \Rightarrow x^2 = e^6$.

$$x^2 = e^6$$

Raise e to both sides.

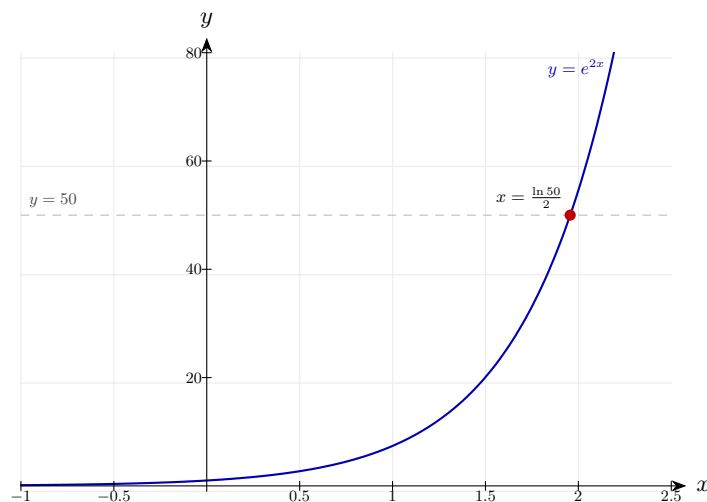
Step 4: Take both square roots.

$$x = \pm e^3$$

Both positive and negative values are valid since $\ln(x^2)$ is defined for $x \neq 0$.

Final answer: (a) $x = \frac{\ln 50}{2} \approx 1.96$. (b) $x = \pm e^3$.

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 35 (e12-035)

Difficulty: Foundation / Marks: 8

(a) Given that $\log_5 x = p$ and $\log_5 y = 4p$, write y in terms of x .

(b) Given that $\ln x = 2$ and $\ln y = -4$, write y in terms of x .

(c) Solve $e^{3x-1} = 8$, giving an exact answer.

Worked solution.

Step 1: Part (a): From $\log_5 x = p$: $x = 5^p \Rightarrow 5 = x^{1/p}$.

$$5 = x^{1/p}$$

Express the base in terms of x .

Step 2: From $\log_5 y = 4p$: $y = 5^{4p}$. Substitute $5 = x^{1/p}$.

$$y = \left(x^{1/p}\right)^{4p} = x^4$$

$$\frac{4p}{p} = 4.$$

Step 3: Part (b): From $\ln x = 2$: $x = e^2 \Rightarrow e = x^{1/2}$.

$$e = x^{1/2} = \sqrt{x}$$

Raise both sides of $x = e^2$ to $\frac{1}{2}$.

Step 4: From $\ln y = -4$: $y = e^{-4}$. Substitute.

$$y = \left(x^{1/2}\right)^{-4} = x^{-2} = \frac{1}{x^2}$$

$$\frac{-4}{2} = -2.$$

Step 5: Part (c): Apply $\ln$ to both sides.

$$3x - 1 = \ln 8$$

$$\ln e^{3x-1} = 3x - 1.$$

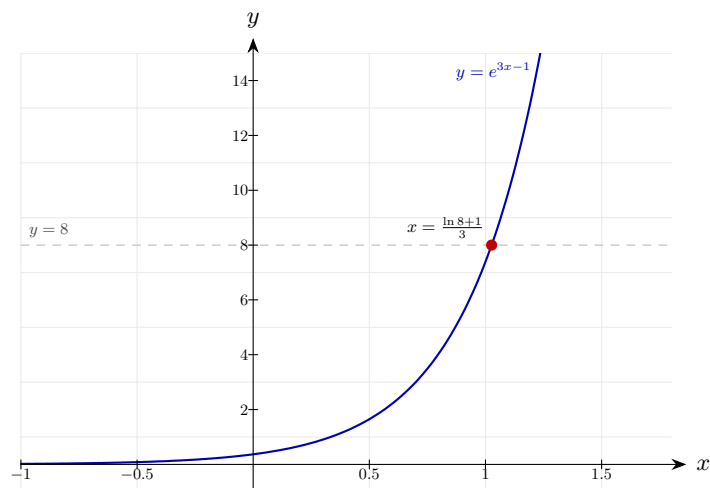
Step 6: Solve for x .

$$3x = \ln 8 + 1 \Rightarrow x = \frac{\ln 8 + 1}{3}$$

Exact answer. Note: $\ln 8 = 3 \ln 2$, so $x = \frac{3 \ln 2 + 1}{3} = \ln 2 + \frac{1}{3}$.

Final answer: (a) $y = x^4$. (b) $y = \frac{1}{x^2}$. (c) $x = \frac{\ln 8 + 1}{3}$.

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 36 (e12-036)

Difficulty: Foundation / Marks: 1

Evaluate $\log_2 32$.

Worked solution.

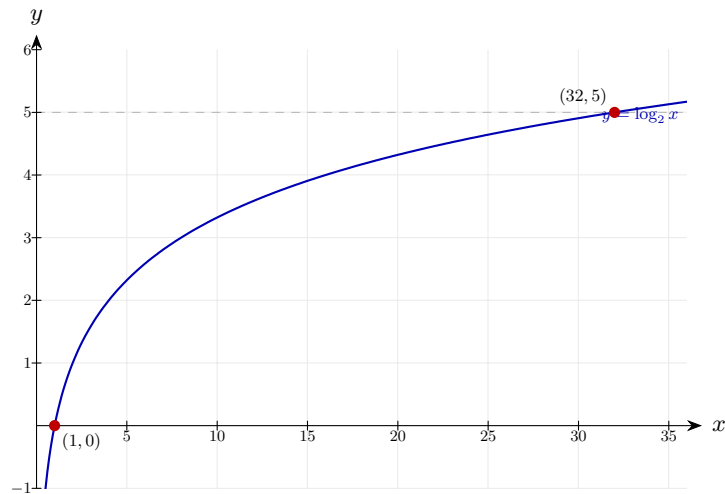
Find the power

$$2^5 = 32 \implies \log_2 32 = 5$$

By definition, $\log_a x = y \Leftrightarrow a^y = x$, so we need the power of 2 that gives 32. Since $2^5 = 32$, the logarithm is 5.

Final answer: 5

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 37 (e12-037)

Difficulty: Foundation / Marks: 1

Evaluate $\log_3 81$.

Worked solution.

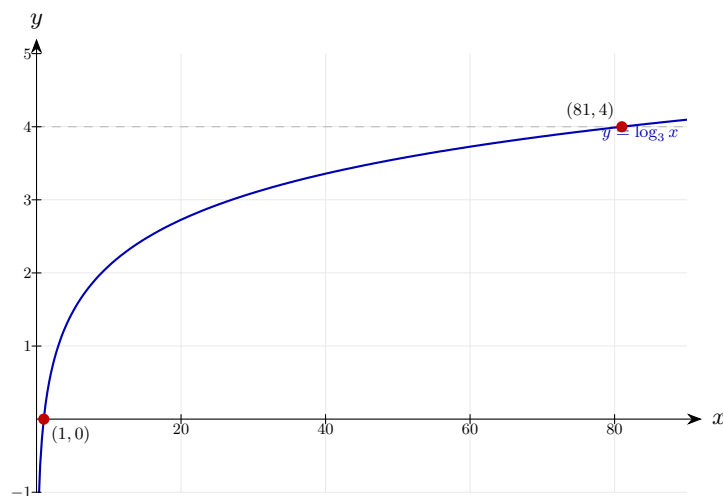
Find the power

$$3^4 = 81 \implies \log_3 81 = 4$$

$\log_3 81$ asks: what power of 3 gives 81? Listing powers $3^1 = 3$, $3^2 = 9$, $3^3 = 27$, $3^4 = 81$ shows the answer is 4.

Final answer: 4

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 38 (e12-038)

Difficulty: Foundation / Marks: 2

Evaluate $\log_{10} 0.001$.

Worked solution.

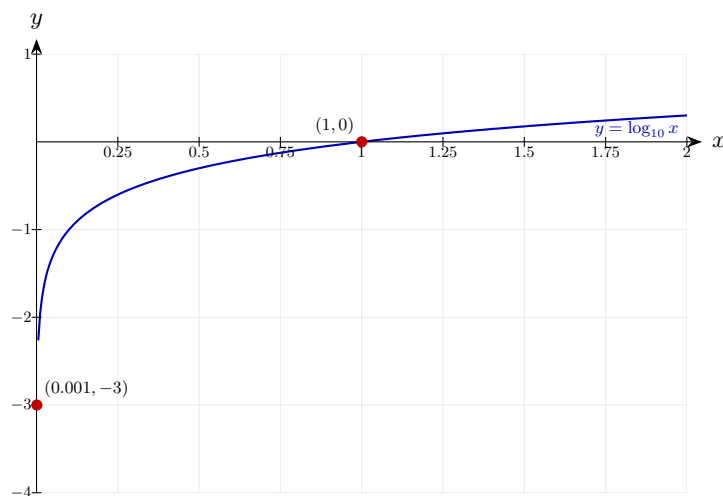
Write as power of 10

$$0.001 = 10^{-3} \implies \log_{10} 0.001 = -3$$

Write the argument as a power of the base: $0.001 = \frac{1}{1000} = 10^{-3}$. A negative exponent gives a value between 0 and 1, so the logarithm is negative.

Final answer: -3

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 39 (e12-039)

Difficulty: Foundation / Marks: 2

Evaluate $\log_5 \frac{1}{25}$.

Worked solution.

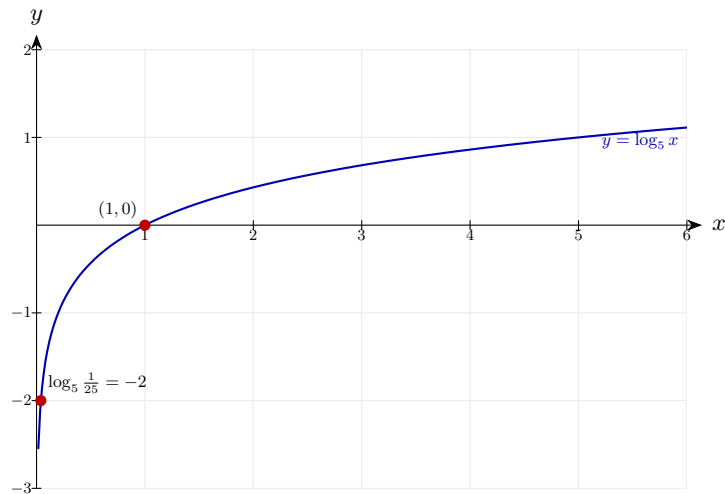
Write as power

$$\frac{1}{25} = 5^{-2} \implies \log_5 \frac{1}{25} = -2$$

Rewrite the argument as a power of 5: $\frac{1}{25} = \frac{1}{5^2} = 5^{-2}$. Reciprocals of the base give negative logarithms.

Final answer: -2

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 40 (e12-040)

Difficulty: Foundation / Marks: 1

Write $3^4 = 81$ in logarithmic form.

Worked solution.

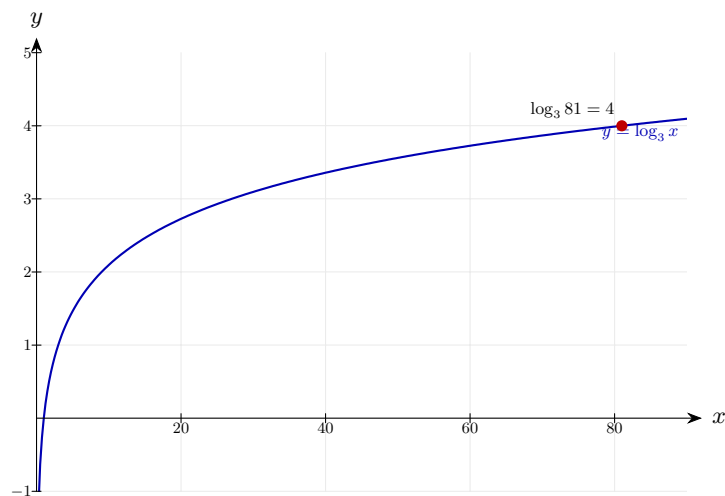
Convert

$$\log_3 81 = 4$$

Apply the definition $a^c = b \Leftrightarrow \log_a b = c$: the base of the power (3) becomes the base of the log, and the exponent (4) becomes the value of the log.

Final answer: $\log_3 81 = 4$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 41 (e12-041)

Difficulty: Foundation / Marks: 1

Write $\log_2 64 = 6$ in exponential form.

Worked solution.

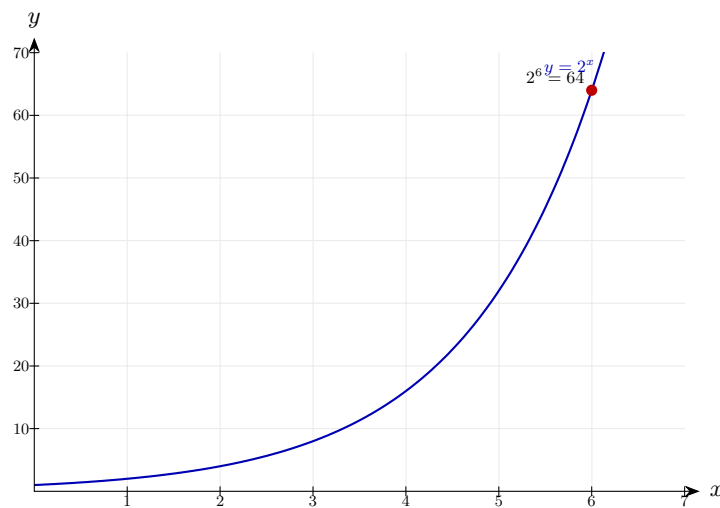
Convert

$$2^6 = 64$$

Reverse the log definition: $\log_a b = c \Leftrightarrow a^c = b$. The base of the log (2) becomes the base of the power, and the value of the log (6) becomes the exponent.

Final answer: $2^6 = 64$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 42 (e12-042)

Difficulty: Foundation / Marks: 1

Evaluate $\log_4 1$.

Worked solution.

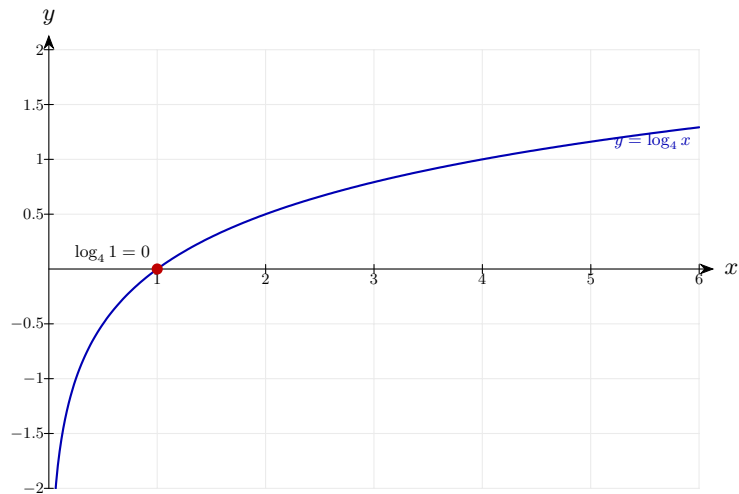
Any base to power 0 is 1

$$4^0 = 1 \implies \log_4 1 = 0$$

This is the key fact $\log_a 1 = 0$ for every valid base, because $a^0 = 1$ for any $a > 0$.

Final answer: 0

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 43 (e12-043)

Difficulty: Foundation / Marks: 1

Evaluate $\log_7 7$.

Worked solution.

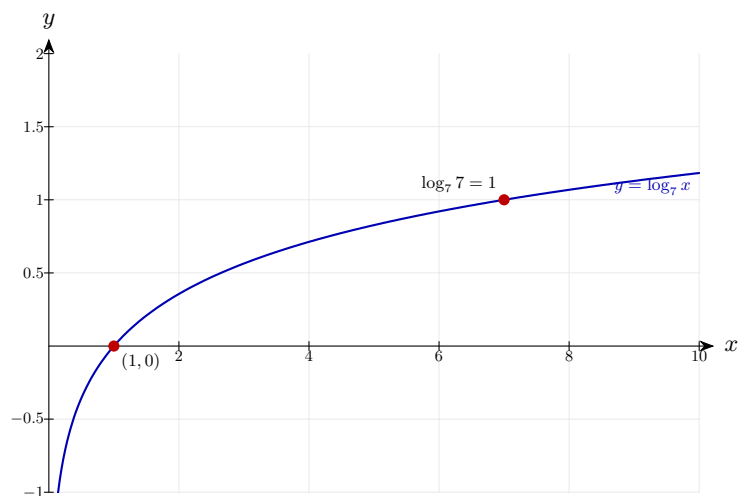
$$\log_a a = 1$$

$$7^1 = 7 \implies \log_7 7 = 1$$

Another key fact: $\log_a a = 1$ for any valid base, because raising the base to the power 1 just gives the base back.

Final answer: 1

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 44 (e12-044)

Difficulty: Foundation / Marks: 2

Solve $\log_2 x = 5$.

Worked solution.

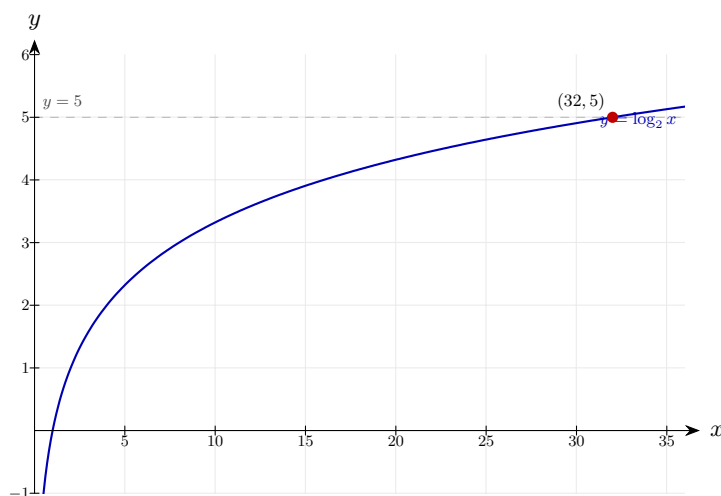
Convert to exponential

$$x = 2^5 = 32$$

Use $\log_a x = y \Leftrightarrow a^y = x$ to rewrite the equation as a power. Here $x = 2^5 = 32$.

Final answer: $x = 32$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 45 (e12-045)

Difficulty: Foundation / Marks: 2

Solve $\log_3 x = -2$.

Worked solution.

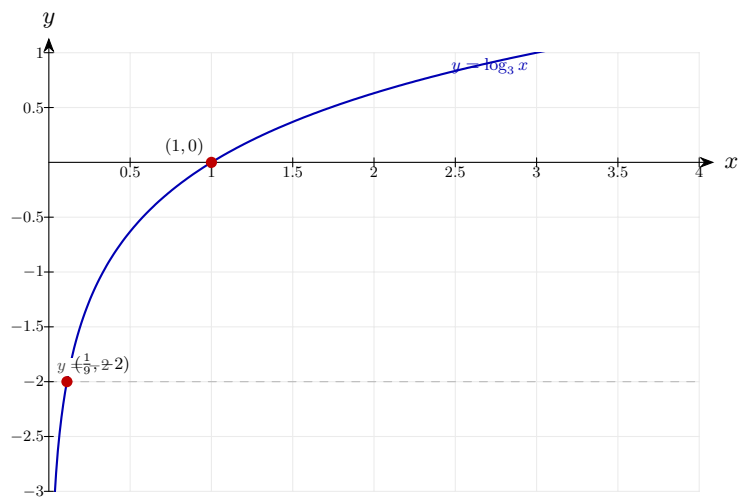
Convert to exponential

$$x = 3^{-2} = \frac{1}{9}$$

Apply the definition $\log_a x = y \Leftrightarrow a^y = x$. A negative log means x is a reciprocal: $3^{-2} = \frac{1}{3^2} = \frac{1}{9}$.

Final answer: $x = 1/9$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 46 (e12-046)

Difficulty: Foundation / Marks: 2

Solve $\log_x 49 = 2$.

Worked solution.

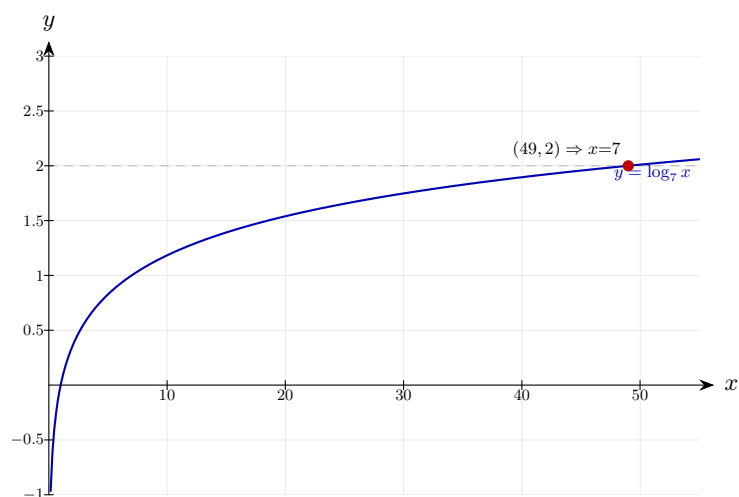
Convert

$$x^2 = 49 \implies x = 7$$

Positive base only.

Final answer: $x = 7$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 47 (e12-047)

Difficulty: Foundation / Marks: 1

Evaluate $\ln e^4$.

Worked solution.

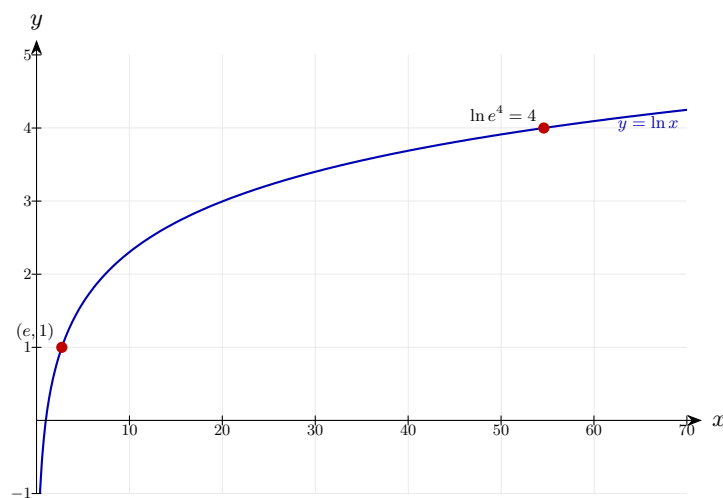
$\ln$ and e are inverses

$$\ln e^4 = 4$$

Since $\ln$ is $\log_e$, the inverse property $\log_a(a^x) = x$ gives $\ln(e^4) = 4$ directly.

Final answer: 4

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 48 (e12-048)

Difficulty: Foundation / Marks: 1

Evaluate $e^{\ln 3}$.

Worked solution.

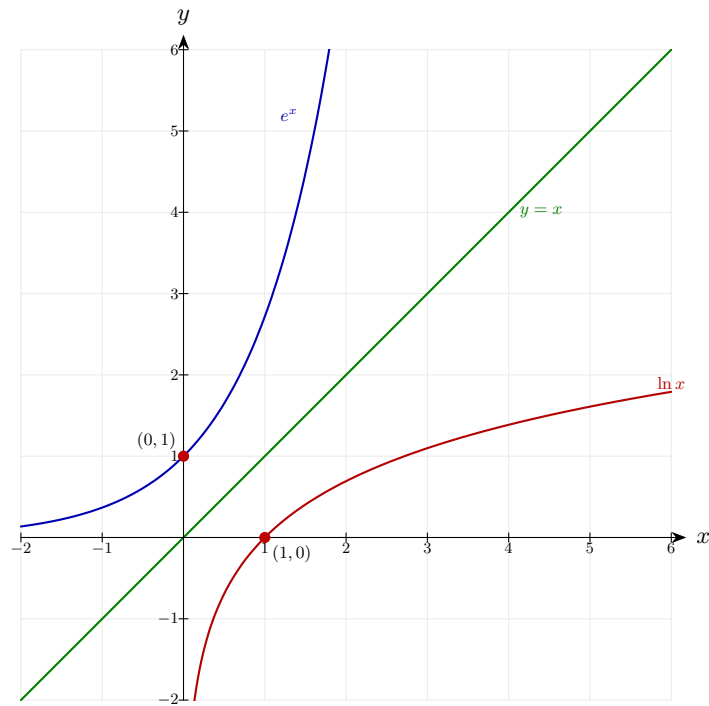
e and $\ln$ are inverses

$$e^{\ln 3} = 3$$

The exponential and natural logarithm undo each other: $e^{\ln x} = x$ for any $x > 0$. (Recall that $\ln$ is only defined for positive arguments.)

Final answer: 3

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 49 (e12-049)

Difficulty: Foundation / Marks: 2

Solve $\ln x = 2$.

Worked solution.

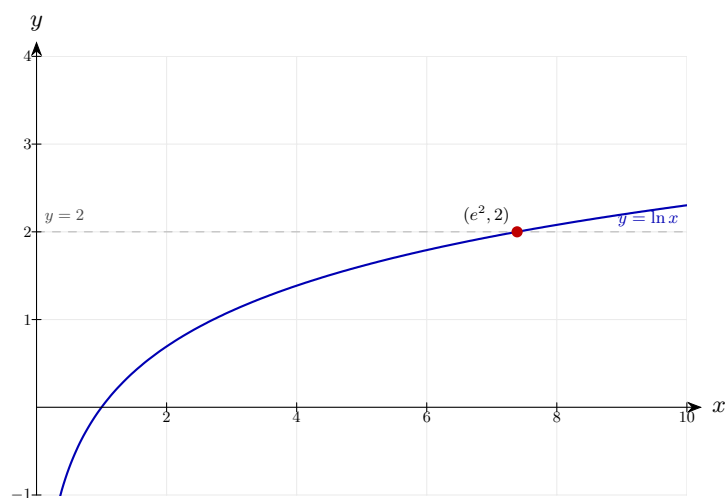
Exponentiate

$$x = e^2 \approx 7.39$$

Raise e to both sides to undo the natural log: $e^{\ln x} = x$, so $x = e^2$. Numerically $e^2 \approx 7.389$.

Final answer: $x = e^2$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 50 (e12-050)

Difficulty: Foundation / Marks: 3

Solve $2^x = 10$. Give your answer to 3 s.f.

Worked solution.

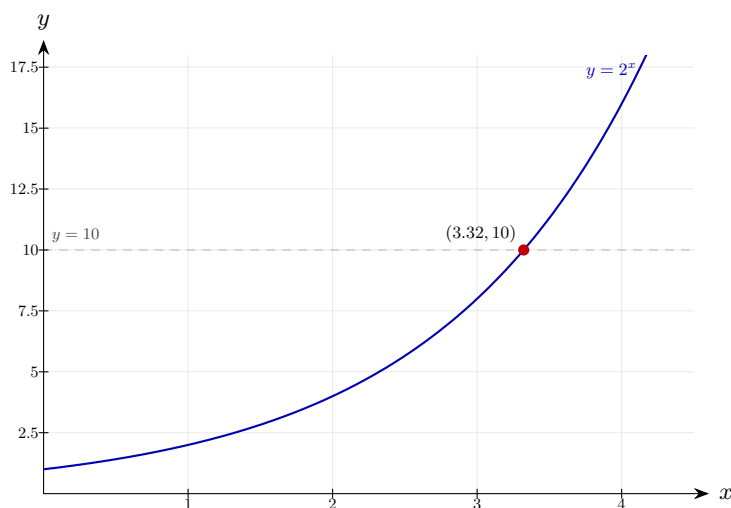
Take logs

$$x \ln 2 = \ln 10 \implies x = \frac{\ln 10}{\ln 2} = 3.32$$

Take $\ln$ of both sides and use the power law $\ln(a^x) = x \ln a$ to bring x down. Dividing gives the change-of-base form $x = \frac{\ln 10}{\ln 2} = \log_2 10$ — change of base is exactly what lets you compute a logarithm with an awkward base on a calculator that only has $\ln$ or $\log_{10}$.

Final answer: $x = 3.32$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 51 (e12-051)

Difficulty: Foundation / Marks: 3

Solve $5^x = 20$. Give your answer to 3 s.f.

Worked solution.

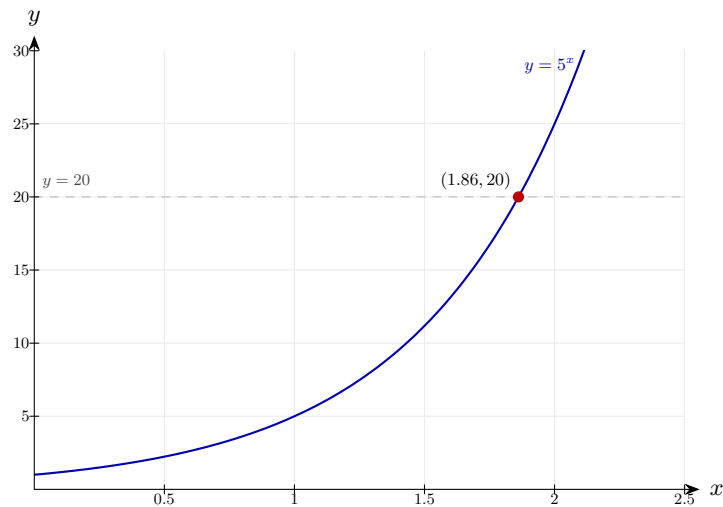
Take logs

$$x = \frac{\ln 20}{\ln 5} = \frac{3.00}{1.609} = 1.86$$

Take $\ln$ of both sides, then use $\ln(5^x) = x \ln 5$ and divide. Note $\ln 20 \approx 2.996$, shown here rounded to 3.00 (3 s.f.) — keep extra precision in intermediate steps to avoid rounding error in the final answer.

Final answer: $x = 1.86$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 52 (e12-052)

Difficulty: Foundation / Marks: 4

Solve $3^{2x-1} = 15$. Give your answer to 3 s.f.

Worked solution.

Step 1: Take logs

$$(2x - 1) \ln 3 = \ln 15$$

Take $\ln$ of both sides. The power law $\ln(a^k) = k \ln a$ brings the whole exponent $2x - 1$ down as a factor — treat it as a single bracket to keep the algebra clean.

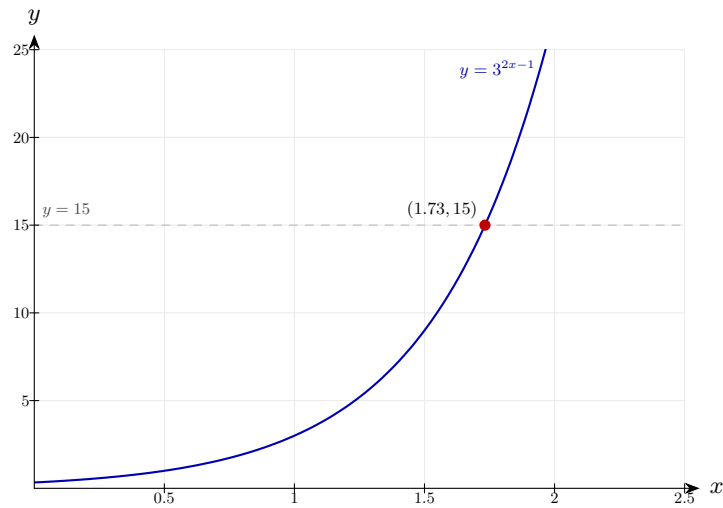
Step 2: Solve

$$2x - 1 = \frac{\ln 15}{\ln 3} = 2.465 \implies x = 1.73$$

Divide by $\ln 3$ to isolate the linear expression in x , then rearrange. Equivalently $\frac{\ln 15}{\ln 3} = \log_3 15$ by change of base.

Final answer: $x = 1.73$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 53 (e12-053)

Difficulty: Foundation / Marks: 2

Evaluate $\log_8 2$.

Worked solution.

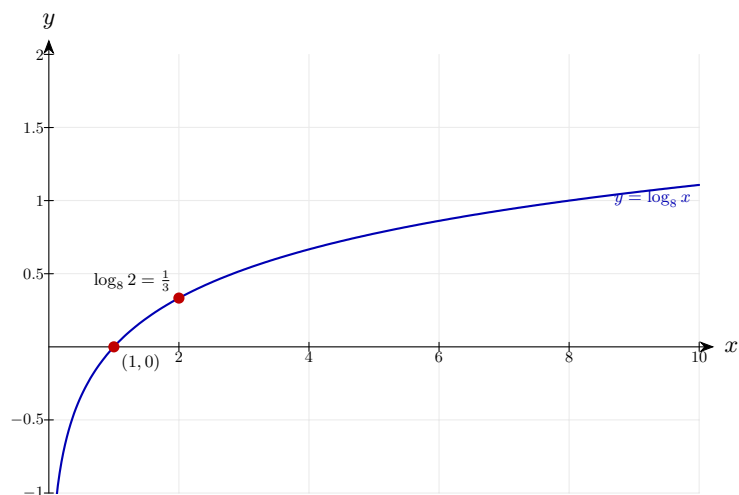
Write 2 as power of 8

$$8^{1/3} = 2 \implies \log_8 2 = \frac{1}{3}$$

Ask what power of 8 gives 2. Since $2 = \sqrt[3]{8} = 8^{1/3}$, the log equals $\frac{1}{3}$. Fractional logs always come from roots of the base.

Final answer: $1/3$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 54 (e12-054)

Difficulty: Foundation / Marks: 2

Evaluate $\log_9 3$.

Worked solution.

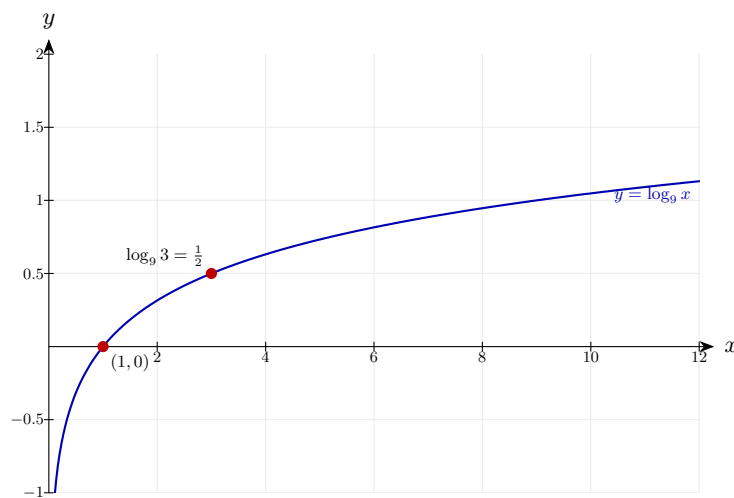
Write 3 as power of 9

$$9^{1/2} = 3 \implies \log_9 3 = \frac{1}{2}$$

A power of $\frac{1}{2}$ is a square root, and $\sqrt{9} = 3$. So by the log definition, $\log_9 3 = \frac{1}{2}$.

Final answer: $1/2$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 55 (e12-055)

Difficulty: Foundation / Marks: 3

Sketch $y = \ln x$. State the x-intercept and asymptote.

Worked solution.

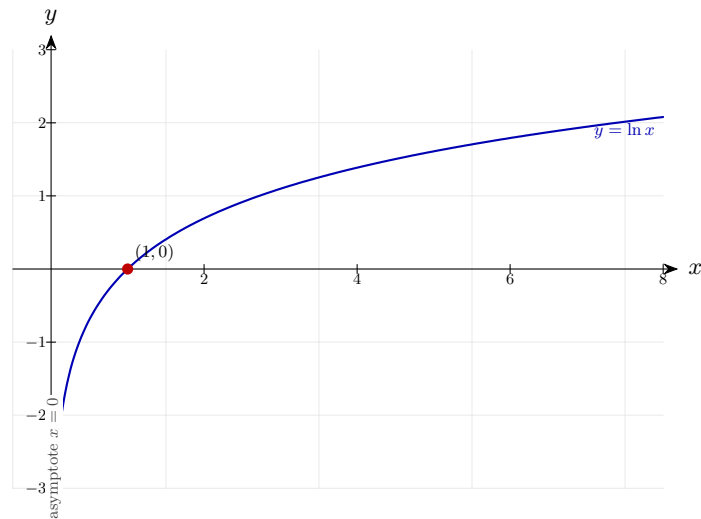
Key features

x-int: (1,0); asymptote: $x = 0$

Defined for $x > 0$ only.

Final answer: x-int (1,0); asymptote $x = 0$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 56 (e12-056)

Difficulty: Foundation / Marks: 3

Sketch $y = \log_{10} x$ and $y = 10^x$ on the same axes. What is the relationship?

Worked solution.

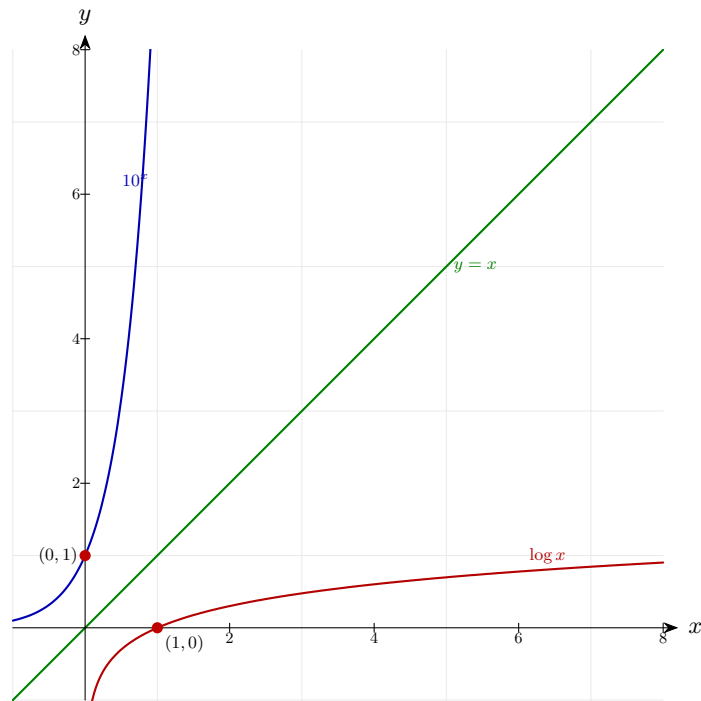
Inverse functions

Reflections of each other in $y = x$

$y = \log_{10} x$ and $y = 10^x$ are inverse functions, so their graphs are mirror images of each other in the line $y = x$. The exponential passes through $(0, 1)$; the log passes through $(1, 0)$ — these intercepts are reflections of each other.

Final answer: Inverse functions; reflections in $y = x$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 57 (e12-057)

Difficulty: Foundation / Marks: 3

Solve $e^{2x} = 8$. Give exact answer.

Worked solution.

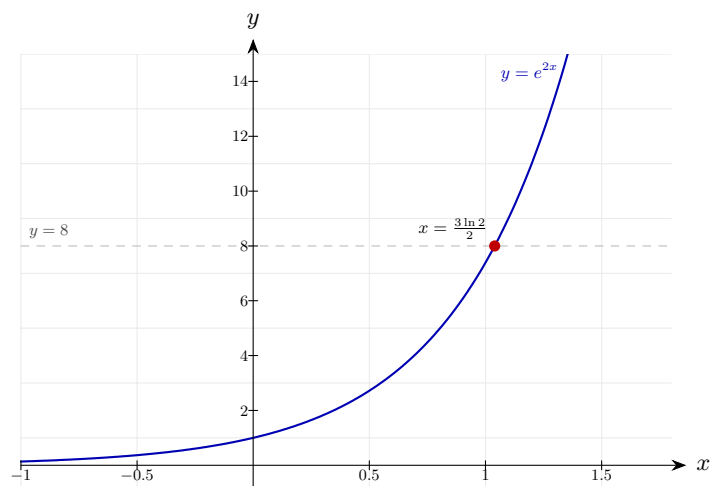
Take $\ln$

$$2x = \ln 8 = 3 \ln 2 \implies x = \frac{3 \ln 2}{2}$$

Natural log appears naturally because the base is e : taking $\ln$ of both sides uses the inverse property $\ln(e^{2x}) = 2x$. Then $\ln 8 = \ln(2^3) = 3 \ln 2$ by the power law, giving the tidiest exact form.

Final answer: $x = (3 \ln 2)/2$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 58 (e12-058)

Difficulty: Foundation / Marks: 3

Solve $\ln(2x + 1) = 3$.

Worked solution.

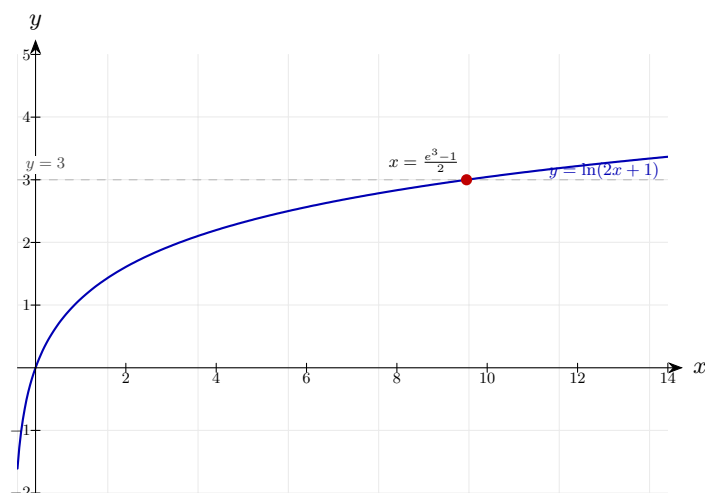
Exponentiate

$$2x + 1 = e^3 \implies x = \frac{e^3 - 1}{2}$$

Raise e to both sides to undo the $\ln$: $e^{\ln(2x+1)} = 2x + 1$. Then solve the resulting linear equation. (Check: $2x + 1 = e^3 > 0$, so the original $\ln$ is defined.)

Final answer: $x = (e^3 - 1)/2$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 59 (e12-059)

Difficulty: Foundation / Marks: 4

Solve $\log_2(x + 3) + \log_2(x - 1) = 3$.

Worked solution.

Step 1: Combine logs

$$\log_2[(x + 3)(x - 1)] = 3$$

Use the addition law $\log_a P + \log_a Q = \log_a(PQ)$ to combine into a single log. Both arguments must be positive, so we need $x > 1$ — keep this constraint in mind when picking valid roots.

Step 2: Convert

$$(x + 3)(x - 1) = 8 \implies x^2 + 2x - 11 = 0$$

Convert to index form using $\log_a M = c \Leftrightarrow M = a^c$, so $(x+3)(x-1) = 2^3 = 8$. Expand and rearrange to a quadratic.

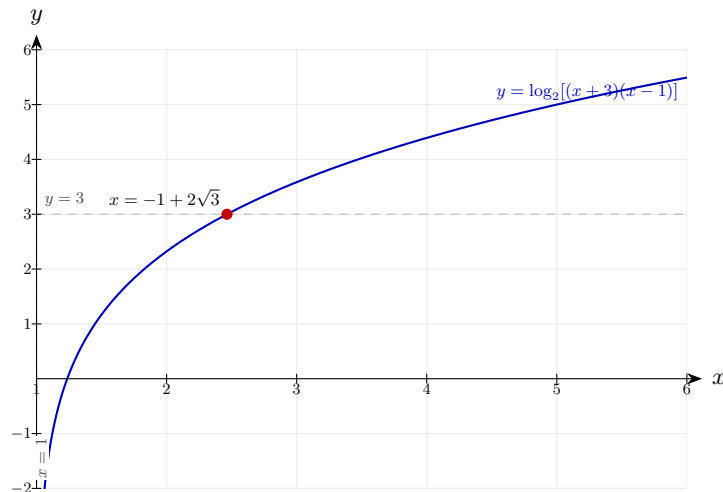
Step 3: Solve

$$x = \frac{-2 + \sqrt{48}}{2} = -1 + 2\sqrt{3} \approx 2.46$$

Reject negative root (need $x > 1$).

Final answer: $x = -1 + 2\sqrt{3}$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 60 (e12-060)

Difficulty: Foundation / Marks: 1

Simplify $\log_a a^5$.

Worked solution.

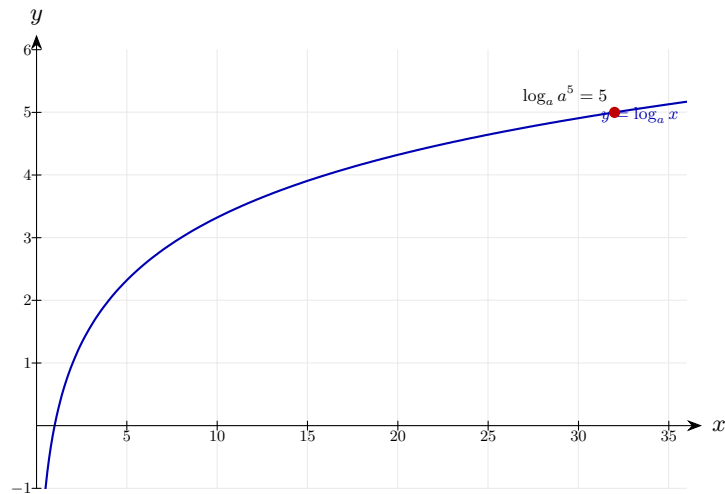
Power rule

$$\log_a a^5 = 5$$

Apply the inverse property $\log_a(a^x) = x$. Equivalently, by the power law $\log_a(a^5) = 5 \log_a a = 5 \times 1 = 5$.

Final answer: 5

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 61 (e12-061)

Difficulty: Foundation / Marks: 2

If $\log_3 x = 2$ and $\log_3 y = 4$, find $\log_3(xy)$.

Worked solution.

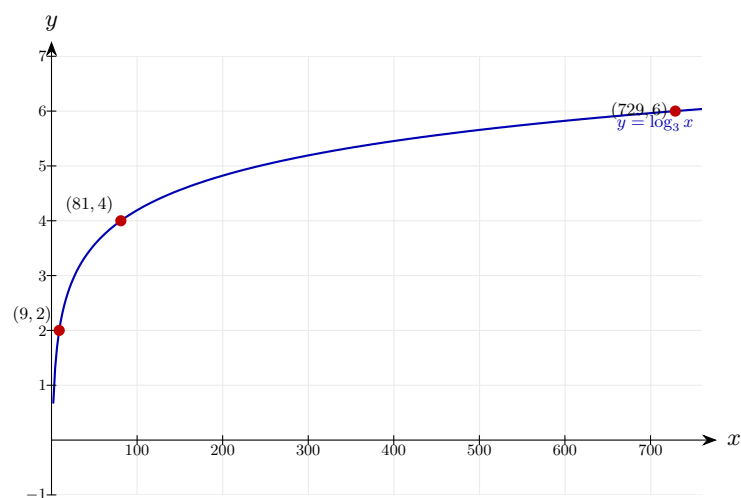
Addition law

$$\log_3(xy) = \log_3 x + \log_3 y = 2 + 4 = 6$$

The log of a product is the sum of the logs: $\log_a(xy) = \log_a x + \log_a y$. This works only because both logs share the same base.

Final answer: 6

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 62 (e12-062)

Difficulty: Foundation / Marks: 3

Solve $10^{x-1} = 500$. Give answer to 3 s.f.

Worked solution.

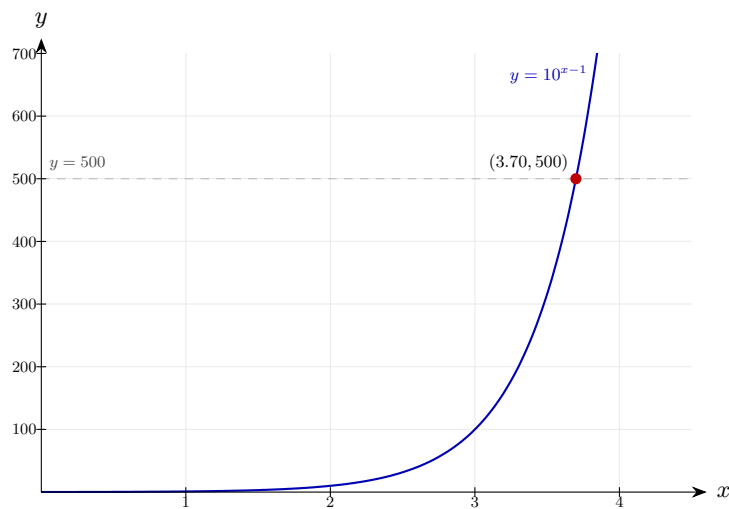
Take $\log 10$

$$x - 1 = \log_{10} 500 = 2.699 \implies x = 3.70$$

Base-10 log is the natural choice when the base of the power is 10, because $\log_{10}(10^{x-1}) = x - 1$ directly. Then add 1 to find x .

Final answer: $x = 3.70$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 63 (e12-063)

Difficulty: Foundation / Marks: 2

Express $2 \log a + 3 \log b$ as a single logarithm.

Worked solution.

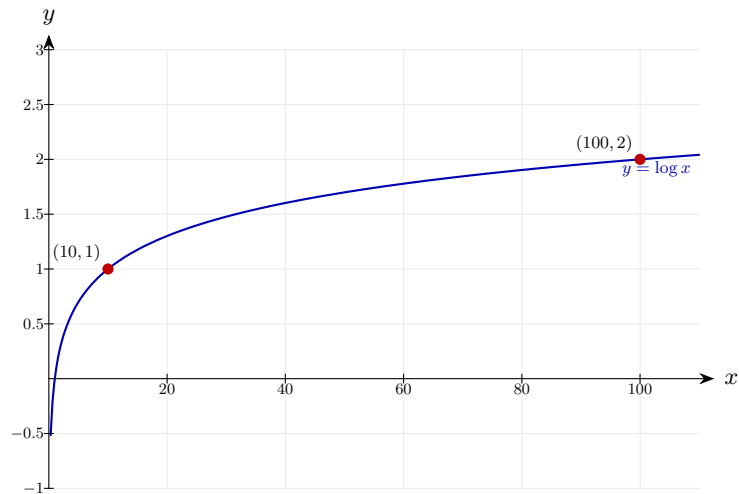
Power then product

$$\log a^2 + \log b^3 = \log(a^2 b^3)$$

First use the power law $k \log a = \log(a^k)$ to move each coefficient inside. Then combine using the addition law $\log P + \log Q = \log(PQ)$.

Final answer: $\log(a^2 b^3)$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 64 (e12-064)

Difficulty: Foundation / Marks: 3

Express $\log 12 - \log 4 + \log 3$ as a single logarithm and evaluate.

Worked solution.

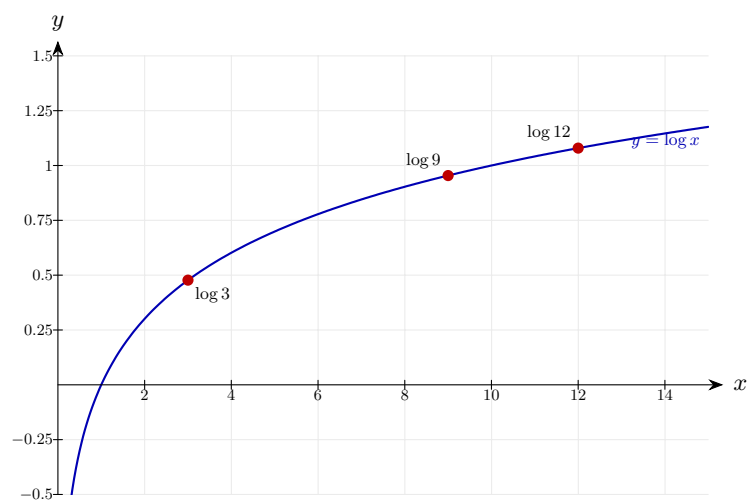
Combine

$$\log \frac{12 \times 3}{4} = \log 9$$

Apply the addition law to the $+\log$ terms (numerator) and the subtraction law to the $-\log$ (denominator): $\log A - \log B + \log C = \log \frac{AC}{B}$. The result simplifies to $\log 9$.

Final answer: $\log 9$

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 65 (e12-065)

Difficulty: Foundation / Marks: 3

Solve $\log_5 x + \log_5 4 = 2$.

Worked solution.

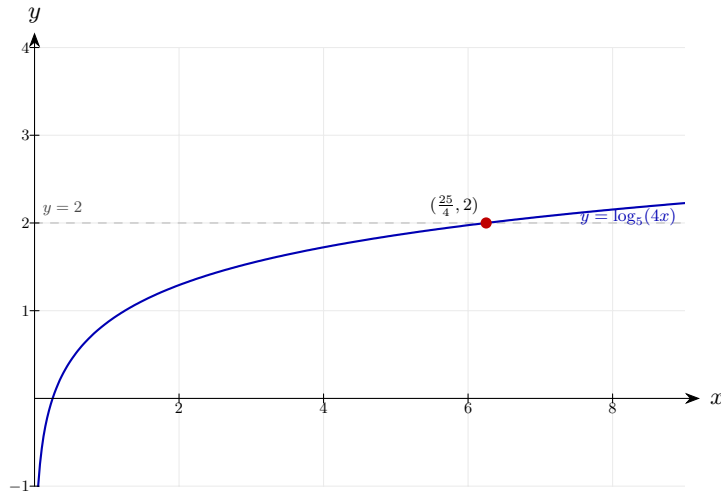
Combine

$$\log_5(4x) = 2 \implies 4x = 25 \implies x = \frac{25}{4}$$

Combine the two logs using the addition law, then convert to index form $\log_5 M = 2 \Leftrightarrow M = 5^2 = 25$.
(Check $x > 0$: $\frac{25}{4} > 0$, so the original $\log_5 x$ is defined.)

Final answer: $x = 25/4$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 66 (e12-066)

Difficulty: Foundation / Marks: 3

Given $\log_2 3 = 1.585$, find $\log_2 12$.

Worked solution.

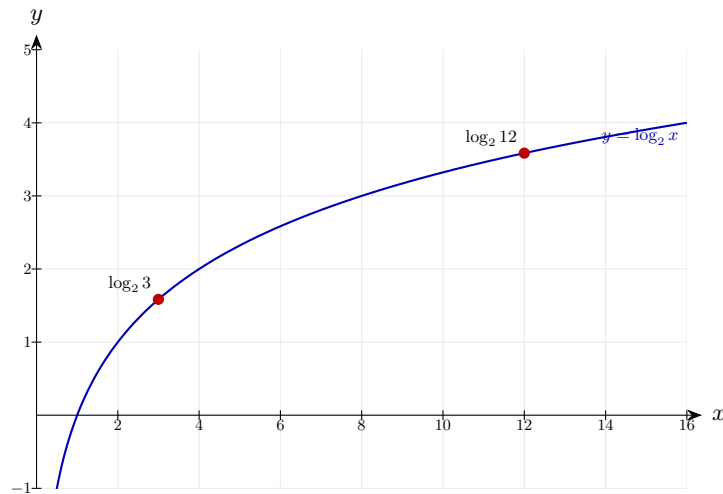
Write $12 = 4 \times 3$

$$\log_2 12 = \log_2 4 + \log_2 3 = 2 + 1.585 = 3.585$$

Factorise the argument so each piece is easy: $12 = 4 \times 3$, then split via the addition law. $\log_2 4 = 2$ is exact, and $\log_2 3$ is the given value.

Final answer: 3.585

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 67 (e12-067)

Difficulty: Foundation / Marks: 5

Solve $e^x + e^{-x} = 3$.

Worked solution.

Step 1: Let $u = e^x$

$$u + \frac{1}{u} = 3 \implies u^2 - 3u + 1 = 0$$

Substitute $u = e^x$; then $e^{-x} = \frac{1}{u}$. Multiplying through by u (which is positive since $e^x > 0$, so no extraneous roots) clears the fraction and gives a quadratic in u .

Step 2: Solve

$$u = \frac{3 \pm \sqrt{5}}{2}$$

Apply the quadratic formula. Both roots are positive (since $\sqrt{5} < 3$), so both are valid values for $u = e^x$ — important because $\ln$ is only defined for positive arguments.

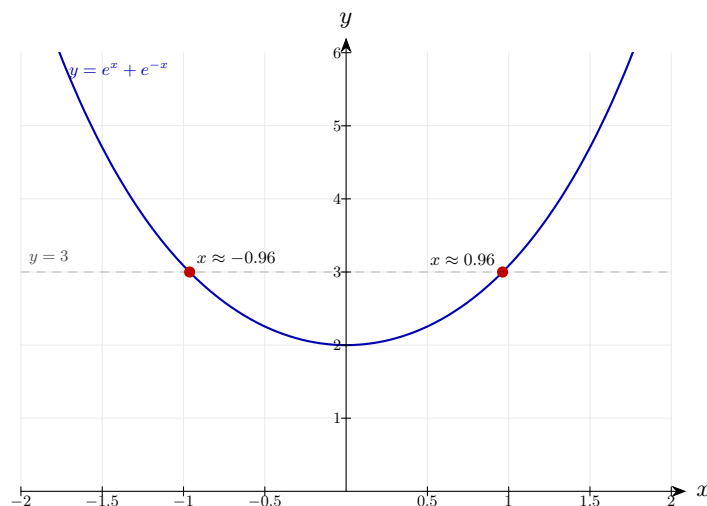
Step 3: Take $\ln$

$$x = \ln\left(\frac{3 + \sqrt{5}}{2}\right) \text{ or } x = \ln\left(\frac{3 - \sqrt{5}}{2}\right)$$

Both positive so both valid.

Final answer: $x = \ln((3 + \sqrt{5})/2)$ or $x = \ln((3 - \sqrt{5})/2)$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 68 (e12-068)

Difficulty: Foundation / Marks: 3

The Richter scale: an earthquake of magnitude M has energy $E = 10^{1.5M+4.8}$ joules. Find M when $E = 10^{12}$.

Worked solution.

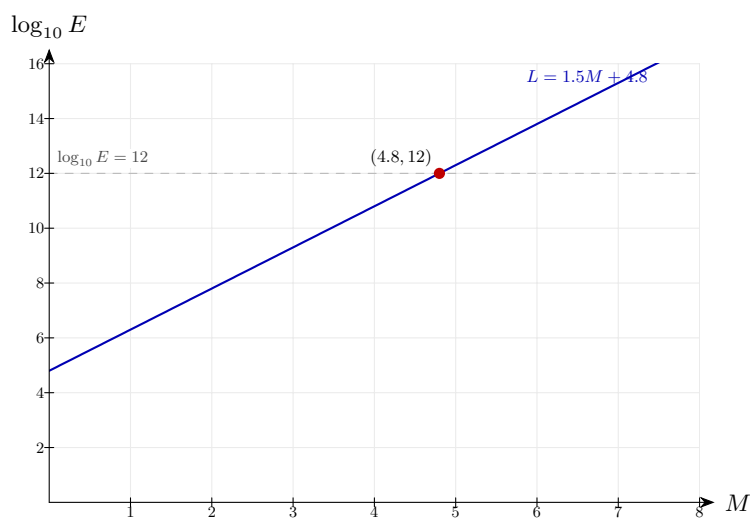
Set equal

$$1.5M + 4.8 = 12 \implies 1.5M = 7.2 \implies M = 4.8$$

Since both sides are powers of 10 with the same base, the exponents must be equal: $10^a = 10^b \implies a = b$. This avoids needing logs at all when both sides share a base.

Final answer: $M = 4.8$

Diagram.



The logarithmic curve plotted with key points marked, and the dashed line through the solution.

Logarithms 69 (e12-069)

Difficulty: Foundation / Marks: 2

Use the change of base formula to evaluate $\log_3 20$ to 3 s.f.

Worked solution.

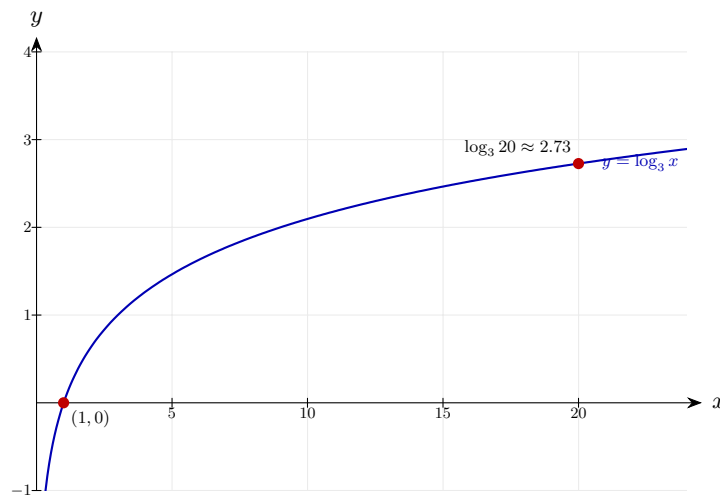
Change of base

$$\log_3 20 = \frac{\ln 20}{\ln 3} = \frac{3.00}{1.099} = 2.73$$

Change of base: $\log_a x = \frac{\ln x}{\ln a}$ (or any other common base such as $\log_{10}$). This is essential because most calculators only have $\ln$ and $\log_{10}$ buttons, not arbitrary bases.

Final answer: 2.73

Diagram.



The logarithmic curve plotted with key points marked.

Logarithms 70 (e12-070)

Difficulty: Foundation / Marks: 5

Solve $\log_2 x + \log_4 x = 6$. [Hint: use change of base.]

Worked solution.

Step 1: Change base of $\log_4 x$

$$\log_4 x = \frac{\log_2 x}{\log_2 4} = \frac{\log_2 x}{2}$$

The two logs have different bases, so we cannot combine them directly. Use change of base $\log_a x = \frac{\log_b x}{\log_b a}$ with $b = 2$ to rewrite $\log_4 x$ in terms of $\log_2 x$; since $\log_2 4 = 2$, this halves it.

Step 2: Substitute

$$\log_2 x + \frac{\log_2 x}{2} = 6 \implies \frac{3}{2} \log_2 x = 6$$

Both terms are now in the same base, so collect them. Treating $\log_2 x$ as a single unknown, $1 + \frac{1}{2} = \frac{3}{2}$.

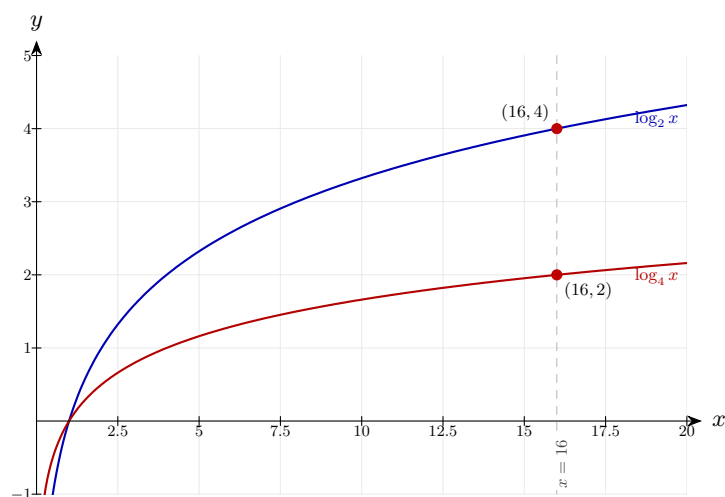
Step 3: Solve

$$\log_2 x = 4 \implies x = 16$$

Multiply both sides by $\frac{2}{3}$ to isolate $\log_2 x$, then convert to index form $x = 2^4 = 16$.

Final answer: $x = 16$

Diagram.



The logarithmic curve plotted with key points marked.
